Spatial Causal Inference for Environmental Data

ENVR Workshop 2026

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https://tinyurl.com/causalsc

October 8, 2026



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- causal inference seems interesting/relevant but I don't know where to start
- have absorbed some causal inference concepts, but never really studied it or had a good explanation of the basics
- I know some of the "spatial confounding" literature in spatial statistics...
  - but I try to read papers in the modern causal inference literature and I don't quite get what it's all about

~~Spatial Causal Inference for Environmental Data~~ An Introduction to Causal Inference for Spatial/Environmental Statisticians

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Workshop Goal

1. Give some orientation of what causal inference actually *is*
   - Intuition that is spread across decades of literature, but missing from modern papers
   - What's so different about methods that are called "causal?"

2. Learn to think about statistics and observational studies from a causal inference perspective
   - Not necessarily restricted to specific methods

3. Outline problems in "spatial causal inference"
   - From a causal inference perspective
   - Spatial Confounding
   - Pointers towards other modern research topics

Workshop is **not** intended to...

- Give a review of all causal inference
- Provide a lot of "how to" implement different methods
- Provide lots of technical detail
- Outline all ongoing work in causal inference for spatial/environmental data

Chapter 1: Introduction to Causal Inference
- Chapter 1.1: What is Causal Inference?
- Chapter 1.2: The Rubin Causal Model

Chapter 2: Estimating Causal Effects
- Chapter 2.1: "Designing" Observational Studies
- Chapter 2.2: Analyzing Observational Studies
- Chapter 2.3: Case Study: Effects of Power Plant Emissions Controls

Chapter 3: Spatial Confounding
- Chapter 3.1: What is Spatial Confounding?
- Chapter 3.2: Spatial Confounding Adjustment
- Chapter 3.3: Case Study: Effects of Power Plant Emissions Controls

Chapter 4: Frontiers in Spatial Causal Inference
- Chapter 4.1: Nonlocal Confounding
- Chapter 4.2: Spatial Interference and Spillover

Different Emphasis of Spatial Statistics and Causal Inference

Broad Generalization

Spatial Statistics

- Complex regression models
- Encode spatial process or correlation
- Models represent mechanistic understanding of data generating process
- Model parameters $\Leftrightarrow$ "effects" of certain features
- Consider how modeling choices impact parameter estimates

Causal Inference

- Formalize potential outcomes, causal effects
- Explicate assumptions needed for causal interpretation
- Identifying assumptions for causal inference
- Statistical model not required

Chapter 1: Introduction to Causal Inference

Chapter 1.1: What is Causal Inference?

Operating Introductory Question:

What is causal inference and how do we mathematically formalize "cause" and "effect"?

or

Correlation is not causation....so what is causation?

Different Meanings of "Causation"

Foundational Reading: (Holland 1986)

1. Ultimate meaningfulness of causation
   - Aristotle!
2. Understanding details of causal mechanisms
   - What is the biological pathway through which inhaled air pollution causes cardiovascular disease?
3. "Causes of effects"
   - What human activities have contributed to climate change?
4. "Effects of causes"
   - How many deaths are prevented by controlling emissions at coal power plants?
   - Statistics has the most to say about this one

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Some Statistical Frameworks and Perspectives

1. **Potential Outcomes Framework**, AKA, Rubin Causal Model, counterfactual framework, Neyman-Rubin Causal Model
2. Causal diagram framework
3. Structural equation modeling
4. Econometrics, "Quasi Experiments"
5. Causal discovery

These are related frameworks. One **goal** of this course is to give a foundation to navigate this literature.

What is Causal Inference Methodology

Causal inference methodology (for today) is a framework for:

1. Carefully defining a causal effect of interest (e.g., of a decision, action, treatment, exposure, policy, etc.)
2. Determining whether/how such an effect can be estimated from observed data

This sequential framing is intentional. Another way to put it is to separate

1. "The Science"
   - Definition of the object of inference, what we wish to learn
2. The Estimation
   - What we can learn about the science
   - How we go about learning it

Effects of *Causes*

In the potential outcomes paradigm . . .

- A **cause** is an *action* or *state* that occurs, relative to some other *action* or *state*
  - E.g., a person can smoke or not smoke
  - E.g., a school can increase teacher salaries or cut teacher pay
  - E.g., air pollution exposure can be high or low
- An **effect** is a consequence of that explicitly defined cause, defined as a comparison between *what would potentially happen under competing states*

Thought Experiment: Does Pollution Cause Heart Attack?

- Martin enrolls in a study because he lives in part of Los Angeles with high air pollution
- We follow Martin for 50 years...
- At age 70, Martin has a heart attack
- Was there a *causal effect* of the high pollution on Martin's heart attack?

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- **Question:** What would we need to know in order to conclude there was a causal effect?

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- We follow Martin for 50 years...
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- Was there a *causal effect* of the high pollution on Martin's heart attack?
- **Question:** What would we need to know in order to conclude there was a causal effect?

- What if we knew the following
  - If pollution had actually been low in Los Angeles for Martin's entire life
  - If Martin would have otherwise lived his life *exactly* the same
  - Martin would **not** have had a heart attack at age 70
- Now would we conclude that the pollution *caused* Martin's heart attack?

Potential Outcomes and Definition of a Causal Effect

In order to know whether a particular action or state *caused* an outcome, we need to know:

1. What *would potentially happen* if the action or state occurred
2. What *would potentially happen* under some alternative action or state

A **causal effect** is a comparison between would potentially happen under competing actions or states:

Outcome(if Action A) vs. Outcome (if Action B)

Foundational Concept : Units

A population of **observational units**, $i \in \mathcal{U}$

- Objects of study on which causes or treatments may act
- Typically a population from which a sample is drawn
  - Humans, households, locations, ...
- Each $i \in \mathcal{U}$ has:
  - *Response variables*, Y_i, of scientific interest
  - *Attributes*, X_i, other variables describing the units
  - No logical distinction, just variables defined on each $i \in \mathcal{U}$

What Statistics *Usually* Does....

Associational Inference

- Write a *statistical model* describing associations between random quantities
- Make inferences about unknown parameters in the model

- *Associational parameters* are determined by $Pr(Y = y, X = x)$
  - E.g., $Pr(Y = y|X = x)$ would describe how Y changes with X
  - Regression $E[Y|X = x]$ is an associational parameter
- "Static" in the sense that associational parameters do not necessarily tell us about what would happen under a change in conditions

Towards Causal Inference

from (Holland 1986)

When can understanding the joint distribution tell us about causal effects?

Causal analysis goes one step further...The additional information needed for making such predictions is provided by **causal assumptions***...one cannot substantiate causal claims from associations alone...behind every causal conclusion there must lie some causal assumption that is* **not testable.**

"causal assumptions" $\Leftrightarrow$ "identifying assumptions"

Foundational Concept : Treatment

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  - *Response variables*, Y_i, of scientific interest
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  - No logical distinction, just variables defined on each $i \in \mathcal{U}$

  - *Treatments*, Z_i, the cause to which each unit is exposed
    - Treatment, exposure, policy,...
    - **Key feature** *could have been different* for a given i
    - Anything that could be reasonably applied or withheld from a unit
    - Think: hypothetical experiment

What Types of "Causes"?

from (Holland 1986)

- Put as bluntly and as contentiously as possible, in this article I take the position that causes are only those things that could, in principle, be treatments in experiments

- I do not intend to limit the discussion to activities within a controlled randomized study. I do it to emphasize...the fact that the effect of a cause is always relative to another cause.

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Implications:

- Comparative questions
  - The effect of a cause is always defined relative to some other cause
- Invite the framing/language of experiments
- Forces us to be extremely explicit about what (mathematically) is meant by "causal effect"

"No causation without manipulation"

Foundational Concept: Potential Outcomes

...to represent faithfully this state of affairs we need not a single variable, Y, to represent a response, but two variables, $Y(Z = z)$ and $Y(Z = z')$...The interpretation of these two values, $Y_i(Z_i = z)$ and $Y_i(Z_i = z')_i$ for a given unit i, is that $Y_i(Z_i = z)$ is the value of the response that would be observed if the unit were exposed to z and $Y_i(Z_i = z')$ is the value that would be observed on the same unit if it were exposed to z' - (Holland 1986) [edited notation]

Key departure from "associational inference"

- $Y_i(Z_i = z)$ the *potential outcome* that would be observed if i had exposure to $Z_i = z$
- $Y_i(Z_i = z')_i$ the *potential outcome* that would be observed if i had exposure to $Z_i = z'$
- Defined:
  - For every value of $Z \in \{z_1, z_2, \ldots, z_T\}$
  - For every $i \in \mathcal{U}$

Potential Outcomes Framing

- $i = 1, 2, \ldots, n$ indexes *units*
- A "treatment" (or "action" or "exposure" or ...) is observed for each i
  - $Z \in \{z_1, z_2, \ldots, z_T\}$ corresponding to different levels of *treatment*
- Observed outcome of interest
  - Y_i, or Y_i^{obs}

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- Observed outcome of interest
  - Y_i, or Y_i^{obs}
- **Potential Outcomes**
  - $Y_i(Z_i = z)$: What *would occur* if $Z_i = z$
  - $Y_i(Z_i = z')$: What *would occur* if $Z_i = z'$
  - $Y_i(Z_i = z'')$ What *would occur* if $Z_i = z''$
  - ...

(need to keep track of all of these)

Notes on Potential Outcomes

$Y_i(Z_i = z)$: What *would occur* of $Z_i = z$
$Y_i(Z_i = z')$: What *would occur* of $Z_i = z'$

Key Idea: All potential outcomes exist

- Innate characteristics that exist before application of treatment
- Z determines which $Y(Z = z)$ becomes observed, but does not affect the values of $(Y(z_1), Y(z_2), \ldots, Y(z_T))$.
- Relationship between potential outcomes and observed data,

$$Y_i = Y_i^{obs} = \sum_{z \in \mathcal{Z}} Y_i(Z_i = z) \times \mathbb{1}(Z_i = z)$$

Foundational Concept: Causal Effect

There is a *causal effect* of Z on Y for unit i if

$$Y_i(Z_i = z') \neq Y_i(Z_i = z)$$

There is a *causal effect* of Z on Y if above holds for any i

Definition **does not** depend on: how many units, which outcome is observed, **parameters in a statistical model**

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The "**fundamental problem of causal inference**:" we only observe a single $Y_i(Z_i)$ corresponding to the observed value of Z_i

A "Statistical Solution" to the Fundamental Problem

- Make use of a *population* of multiple units
- Think about what would happen *on average* under different levels of Z
- Estimate the "other" potential outcome for each unit with the average outcome on units who received the "other" treatment
- Estimate *average causal effects* that average over the units in a sample or population

Basis for estimating average causal effects:

$$E[Y(Z=z)-Y(Z=z')] = E[Y(z)]-E[Y(z')] \stackrel{?}{=} E[Y(z)|Z=z]-E[Y($$

A "Statistical Solution" to the Fundamental Problem

Use average values of units exposed to different levels of Z:

$$\underbrace{E[Y(z) - Y(z')]}_{\text{Causal Estimand}} \stackrel{?}{=} \underbrace{E[Y|Z = z] - E[Y|Z = z']}_{\text{Observed-data Comparison}}$$

- Use units that actually received the "other" treatment
- Are units who received the "other" treatment a good "close substitute" for the unobserved potential outcome
  - Formalize this with *causal identification assumptions*

Chapter 1: Introduction to Causal Inference

Chapter 1.2: The Rubin Causal Model

The Potential Outcomes Framework

Sometimes called the Rubin Causal Model owing to foundational work in Rubin (D. B. Rubin 1974; Donald B. Rubin 1978; D. R. Rubin 1990)

Three main components of the framework:

1. Formulate *potential outcomes* corresponding to various levels of a "treatment"
2. Formulate the *assignment mechanism* governing how treatments are assigned
   - I.e., what was the process that led certain potential outcomes to be *observed* and others not

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Three main components of the framework:

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2. Formulate the *assignment mechanism* governing how treatments are assigned
   - I.e., what was the process that led certain potential outcomes to be *observed* and others not
3. Formulate a (Bayesian) *model* for "the science" represented by all of the potential outcomes and covariates
   - This last part is actually optional...

The Typical Setup

- Data: sample of $i = 1, 2, \ldots, n$ **units** from a target population
- A **treatment** with levels $Z \in \{0, 1\}$
  - Binary Z for simplicity
  - More generally, with T levels
- For each i, two **potential outcomes** exist, $(Y_i(0), Y_i(1))$
- For each i, we observe
  - Z_i denoting treatment of i
  - X_i a vector of pre-treatment covariates
  - At most 1 of the potential outcomes is observed

$$Y_i^{obs} = Y_i(Z_i) = Z_i \cdot Y_i(1) + (1 - Z_i) \cdot Y_i(0)$$
$$Y_i^{mis} = Y_i(1 - Z_i) = (1 - Z_i) \cdot Y_i(1) + Z_i \cdot Y_i(0)$$

A "Statistical Solution" to the Fundamental Problem

Use average values of units exposed to different levels of Z:

$$\underbrace{E[Y_i(1) - Y_i(0)]}_{\text{Causal Estimand}} \stackrel{?}{=} \underbrace{E[Y_i|Z_i = 1] - E[Y_i|Z_i = 0]}_{\text{Observed-data Comparison}}$$

$$\underbrace{E[Y_i(1) - Y_i(0)]}_{\text{Causal Estimand}} \stackrel{?}{=} \underbrace{E\big[E[Y_i|Z_i = 1, X_i = x] - E[Y_i|Z_i = 0, X_i = x]\big]}_{\text{Observed-data Comparison}}$$

- Are units who received the "other" treatment a good "close substitute" for the unobserved potential outcome
  - If **yes** $\rightarrow$ observed-data comparison estimates the causal effect
  - Causal inference formalize what is needed to assume the above equality
    - *Causal identification assumptions*

Causal Estimands
Definitions of Causal Effects

- Functions of potential outcomes define causal effects that we may wish to measure or estimate
- In a statistical sense, these are the estimands - the targets for inference
- Other examples of **causal estimands:**
  - Causal risk ratio: $\frac{Pr(Y(1)=1)}{Pr(Y(0)=1)}$
  - Causal odds ratio: $\frac{Pr(Y(1)=1)/Pr(Y(1)=0)}{Pr(Y(0)=1)/Pr(Y(0)=0)}$
- Could also be defined conditional on some attributes: $E[Y(1)|X = x] - E[Y(0)|X = x]$
- *Identification assumptions* tell us when we can estimate a *causal estimand* with *observed data*

Stable Unit Treatment Value Assumption (SUTVA)

Has been implicit thus far

SUTVA Part I: No "Interference"

The potential outcome for the ith unit does not depend on the treatment assigned to *other* units. Let $\mathbf{Z} = (Z_1, Z_2, \ldots, Z_n)$.

$$Y_i(\mathbf{Z}) = Y_i(Z_i)$$

This is precisely the assumption that gives us (with a binary treatment) only 2 potential outcomes for each unit

Without SUTVA, there could be $>>$ 2 potential outcomes for each i

Settings Where SUTVA May Not Hold

- Infections Diseases
  - *My* infection outcome will depend on whether *my wife* gets treated with vaccine
- Social Networks
  - If an intervention gets *me* to quit smoking, it may induce my *friends* to quit too
- Spatial Environmental Settings
  - Reducing emissions at one point in space may impact pollution at locations downwind

Think of treatment effects that "spill over" across units

Stable Unit Treatment Value Assumption (SUTVA)

Has been implicit thus far

SUTVA Part II: No "Multiple Versions" of Treatment

- AKA, "consistency"

$$Y_i^{obs} = Y_i(z) \quad \text{whenever} \quad Z_i = z$$

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Example Violations:

- $Z \equiv$ "being obese", but there are many different reasons someone might be obese
- $Z \equiv$ "reduction in ambient particulate pollution" but this reduction is achieved by different actions (regulating cars, closing power plants, planting trees)

Multiple versions of treatment $\rightarrow$ poorly defined potential outcomes $\rightarrow$ vague causal questions

The Assignment Mechanism

- **Key** identification assumptions relate to mechanism that dictates which treatments are assigned to which units

  $\Rightarrow$ which potential outcomes are revealed

- That is, the *assignment mechanism* plays a key role in determining which (if any) observed-data comparisons can be interpreted as estimates of causal effects
- Example: Randomized experiments imply certain types of assignment mechanisms

Assignment Mechanism

A probability function,

$$Pr(\mathbf{Z}|\mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1))$$

- Mapping of each possible treatment assignment vector to a probability, given "The Science"
  - I.e., can depend on the unobserved potential outcomes
  - Z is the only random variable!
- Describes the reasons for the missing and observed values of $Y(1)$ and $Y(0)$
- Most key assumptions for causal inference are assumptions about the assignment mechanism!
- In an **experiment** the assignment mechanism is **known**
- In an **observational study** the assignment mechanism is **unknown** $\rightarrow$ assumptions

Observed-Data Comparison $\stackrel{?}{=}$ Causal Effect

from (Gelman, Hill, and Vehtari 2021)

Table: Potential Outcomes ("The Science")

Unit, i	Treatment Z_i	Potential Outcome, $Y_i(0)$	Potential Outcome, $Y_i(1)$
Audrey	0	140	135
Anna	0	140	135
Bob	0	150	140
Bill	0	150	140
Caitlin	1	160	155
Cara	1	160	155
Dave	1	170	160
Doug	1	170	160

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Table: Observed Data

Unit, i	Treatment Z_i	Potential Outcome, $Y_i(0)$	Potential Outcome, $Y_i(1)$	Observed Outcome, Y_i
Audrey	0	140	?	140
Anna	0	140	?	140
Bob	0	150	?	150
Bill	0	150	?	150
Caitlin	1	?	155	155
Cara	1	?	155	155
Dave	1	?	160	160
Doug	1	?	160	160

A "Bad" Assignment Mechanism

- Assigned treatment to those that had systematically higher $Y(1), Y(0)$

$\Rightarrow$ There was some systematic difference between the types of units who were treated vs. not

$\Rightarrow$ Outcomes in the $Z = 0$ group were not a "good substitute" for the unobserved potential outcomes in the $Z = 1$ group

$\Rightarrow$ Observed-data comparison did not reflect the true causal effect

Other "Bad" Assignment Mechanisms

- Doctors treat the sickest patients
- Older people tend to respond to certain advertisements
- Harmful heat exposure is more likely in communities with lower access to medical care
- ... confounding!

In all cases, units with certain Z_i also have certain values of $(Y_i(0), Y_i(1))$:

$\Rightarrow$ Observed data comparisons will misattribute these differences to a treatment effect

Unconfounded Treatment Assignment

- "Bad" assignment mechanisms produce $\mathbf{Z}$ where in $(Y_i(0), Y_i(1))$ differs across treatment groups
  - $\mathbf{Z}$ depends on the values of $(Y_i(0), Y_i(1))$
- This is **confounding**
- But what if the assignment mechanism *didn't* depend on $(Y_i(0), Y_i(1))$?
  - Then the assignment mechanism would be **unconfounded**

Unconfounded Treatment Assignment

An assignment mechanism is *unconfounded* (conditional on $\mathbf{X}$) if it does not depend on *any* potential outcomes

$$Pr(\mathbf{Z}|\mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1)) = Pr(\mathbf{Z}|\mathbf{X})$$

$$Z \perp\!\!\!\perp Y(0), Y(1)|X$$

- Now, we would not expect any systematic imbalance in $(Y(0), Y(1))$ between those who receive $Z = 1$ or 0
- Observed-data comparisons would be "balanced" and thus more capable of estimating causal effects
  - At least on average...

Note: For the purposes of this course, *unconfounded* will sometimes be described as *ignorable*

Observed-Data Comparison $\stackrel{?}{=}$ Causal Effect

from (Gelman, Hill, and Vehtari 2021)

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Example: Hypothetical Dietary Experiment

Observed Data

Table: Observed Data from the Hypothetical Dietary Experiment, **Balanced Design**

Unit, i	Female, x_{1i}	Age, x_{2i}	Treatment Z_i	Potential $Y_i(0)$	Potential $Y_i(1)$
Audrey	1	40	0	140	?
Anna	1	40	1	?	135
Bob	0	50	0	150	?
Bill	0	50	1	?	140
Caitlin	1	60	0	160	?
Cara	1	60	1	?	155
Dave	0	70	0	170	?
Doug	0	70	1	?	160

$(Y(1), Y(0))$ are "balanced" across levels of Z

Achieved by balancing the "right" covariates $X = (x_1, x_2)$

Key Idea: *Balance* Across Treatment Groups

Key Idea: Comparisons between observed outcomes between treated and control must be "balanced" with respect *both* potential outcomes

We can never see these....

- Use pre-treatment covariates as *proxies* for knowledge of both potential outcomes
  - $\rightarrow$ confounding adjustment

- E.g.: If we know that those exposed (i.e., "assigned") to high temperatures
  1. Live in areas with high air pollution
  2. Have higher health risk because of the pollution

$\Rightarrow$ Air pollution is a *confounder* and we should assume it is part of the assignment mechanism

Unconfounded (Ignorable) Study Designs

- Randomized experiments!
  - Randomization $\Rightarrow$ no systematic imbalance in $(Y_i(0), Y_i(0))$
  - No confounding

Unconfounded (Ignorable) Study Designs

- Randomized experiments!
  - Randomization $\Rightarrow$ no systematic imbalance in $(Y_i(0), Y_i(0))$
  - No confounding
- Observational studies
  - Assignment mechanism is unknown
  - Will have to **assume** whether assignment was ignorable
  - That is, assume whether (conditional on $\mathbf{X}$) the treatment is independent of the potential outcomes
  - Treatment is "as though randomized" conditional on $\mathbf{X}$

Ignorable Treatment Assignment

AKA "no unmeasured confounding"

$$Z \perp\!\!\!\perp (Y(0), Y(0))|\mathbf{X}$$

or

$$Pr(\mathbf{Z}|\mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1)) = Pr(\mathbf{Z}|\mathbf{X})$$

$\Rightarrow$ *Within levels of* X, the treatment is "randomly assigned" with respect to the potential outcomes

$\Rightarrow$ Balance across treatment groups, *within levels of* X

- Requires the "right" X
  - Adjustment for the confounders

Conditional Ignorability and Confounding

A *counfounder* is informally defined as a pre-treatment variable that:

1. Is associated with Z
2. Is associated with Y

More formally, *confounders* are covariates required to satisfy

$$Pr(\mathbf{Z}|\mathbf{X}, \mathbf{Y}(1), \mathbf{Y}(0)) = Pr(\mathbf{Z}|\mathbf{X})$$

$$Z \perp\!\!\!\perp Y(0), Y(1)|X$$

Assumption of (Conditional) Ignorability

Causal validity of an observational study will rely heavily on whether the **assumption** of conditional ignorability is plausible

- Goes by many other names
  - Conditional exchangeability, selection on observables, no unmeasured confounders...
- Viability of the assumption relates to what we know (or assume) about the assignment mechanism
- **Cannot be empirically verified**
- Has key (and familiar) implications for estimating causal effects with observed data
- **Not specific to an model!!**

Positivity Assumption

Another assumption about the assignment mechanism

$$0 < Pr(Z_i = 1 | \mathbf{X}_i, Y_i(0), Y_i(1)) < 1$$

- Every unit (i.e., any value of $\mathbf{X}_i$) has positive probability of receiving treatment
- No covariate perfectly predicts treatment
- Related to idea that covariate distributions should "overlap" across treatment levels

References I

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Chapter 2: Estimating Causal Effects

Chapter 2.1: "Designing" Observational Studies

Frame Observational Studies as (Approximate) Randomized Experiments

- Randomized studies are the "gold standard" for estimating causal effects
- Try to relate circumstances of the observational study to those of an experiment
  - Reason about the *assignment mechanism*
  - Think about what an "ideal" experiment *might have* looked like for the problem
- Evaluate: Is the observational study an *approximate* experiment?
  - Is the assignment mechanism unconfounded?
    - Conditional on the right $\mathbf{X}$
  - Are the right $\mathbf{X}$ *balanced*?

Thought Experiment:

Confused? Ask yourself:

If I could have done an experiment in an idealized world, what would that experiment have looked like?

- Can help formalize previous notions, define potential outcomes, treatments, etc.
- Probably not a perfect mapping between your question and an idealized trial, but the exercise is almost always helpful
- Answers should guide how you do the analysis

Thought Experiment:

Confused? Ask yourself:

If I could have done an experiment in an idealized world, what would that experiment have looked like?

- Can help formalize previous notions, define potential outcomes, treatments, etc.
- Probably not a perfect mapping between your question and an idealized trial, but the exercise is almost always helpful
- Answers should guide how you do the analysis

- None of this has anything to do with a statistical model

$\Rightarrow$ Avoids confusing mixing model specifications with definitions of causal effect

5 Steps to Restructure the Data

According to (Gelman, Hill, and Vehtari 2021)

1. Estimands and Confounders
2. Calculate Distance Metrics
3. Restructuring the Data
4. Diagnostics

   (Iterate Steps 2–4 as Necessary)

} "Design Stage"

5. Estimate the treatment effect

} "Analysis Stage"

“Restructure” ⇔ “Design”

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FOR OBJECTIVE CAUSAL INFERENCE, DESIGN TRUMPS ANALYSIS[1]

BY DONALD B. RUBIN

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For obtaining causal inferences that are objective, and therefore have the best chance of revealing scientific truths, carefully designed and executed randomized experiments are generally considered to be the gold standard. Observational studies, in contrast, are generally fraught with problems that compromise any claim for objectivity of the resulting causal inferences. The thesis here is that observational studies have to be carefully designed to approximate randomized experiments, in particular, without examining any final outcome data. Often a candidate data set will have to be rejected as inadequate because of lack of data on key covariates, or because of lack of overlap in the distributions of key covariates between treatment and control groups, often revealed by careful propensity score analyses. Sometimes the template for the approximating randomized experiment will have to be altered, and the use of principal stratification can be helpful in doing this. These issues are discussed and illustrated using the framework of potential outcomes to define causal effects, which greatly clarifies critical issues.

5 Steps to Restructure the Data

According to (Gelman, Hill, and Vehtari 2021)

1. **Estimands and Confounders**
2. Calculate Distance Metrics
3. Restructuring the Data
4. Diagnostics

   (Iterate Steps 2-4 as Necessary)
5. Estimate the treatment effect

What is the Question?

And to whom does it pertain?

Considerations:

- Defining the population of interest
  - Who are the units?
  - Is the "treatment" actually relevant to all of them?
  - Am I interested in all of them?
- Define the treatments being compared
  - What are the "treatment" and "control" conditions?
- Reason about the assignment mechanism
  - Are the key *confounders* available?
- Conceptualize the idealized version of the experiment!

$$\text{Population} + \text{Treatment} \rightarrow \text{Potential outcomes} + \text{Estimand}$$
$$\Rightarrow \text{Precise Question to be Answered}$$

5 Steps to Restructure the Data

According to (Gelman, Hill, and Vehtari 2021)

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2. **Calculate Distance Metrics**
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Measuring SImilarity in $\mathbf{X}$

- Ideal: Units with $Z = 0/1$ that have the *exact same* covariates
  - Can be impractical for many $\mathbf{X}$ with many levels
- Can more generally define a *distance metric* as a function between $\mathbf{X}_i, \mathbf{X}_j$
  - Euclidean distance
  - Mahalanobis distance
  - Propensity score distance
  - ...
- Use distance metric (usually a scalar) to identify units that are "close" in $\mathbf{X}$
  - "Closeness in $\mathbf{X}$" $\approx$ "balance"

Propensity Scores in Observational Studies

$$Pr(\mathbf{Z}|\mathbf{X}) = c\prod_{i=1}^{n} e_i(\mathbf{X}_i)^{Z_i}(1 - e_i(\mathbf{X}_i)^{1-Z_i}$$

- $e_i(\mathbf{X}_i) = e_i = Pr(Z_i = 1|\mathbf{X}_i)$
- Are *unknown* and must be *estimated*
- Represent a "one number" summary of all of the covariate information in $\mathbf{X}_i$
- Specific form of dimension reduction
  - Explicitly defined based on the *assignment mechanism*

  $\Rightarrow$ Confounding adjustment properties
- Closeness in $e_i(\mathbf{X}_i) \approx$ closeness in $\mathbf{X}_i$
  - Use $e_i(\mathbf{X}_i)$ to characterize confounding information

Estimating the Propensity Score (PS)

$$e(\mathbf{X}_i) = Pr(Z_i = 1|\mathbf{X}_i)$$

- $Pr(Z = 1|\mathbf{X}) = E[Z|\mathbf{X}] \rightarrow$ regression for a binary "outocome"
  - Simplest method: Logistic regression
  - Could use any model for binary outcome (GLM, boosted regression, etc.)
  - "Estimated PS": $\hat{e}_i$
- Goal is to estimate $\hat{e}_i$ *that balances covariates,* $\mathbf{X}_i$
- Estimating the PS is just a "means to an end," we don't generally care about interpreting the PS

Main Theoretical Propensity Score Results

from (ROSENBAUM and RUBIN 1983)

1. The propensity score is a *balancing score*
2. Estimates of $e(\mathbf{X})$ can produce sample balance on $\mathbf{X}$
3. Ignorability $|\mathbf{X} \Rightarrow$ Ignorability $|e(\mathbf{X})$

Punchline: If $Z \perp\!\!\!\perp (Y(1), Y(0))|\mathbf{X}$:

$$\underbrace{E[Y_i(1) - Y_i(0)|\hat{e}_i = e]}_{\text{Causal Estimand}} = \underbrace{E[Y_i|\hat{e}_i = e, Z = 1] - E[Y_i|\hat{e}_i = e, Z = 0]}_{\text{Observed Data Comparison}}$$

$\Rightarrow$ *Many* methods for confounding adjustment based on $e(X)$

5 Steps to Restructure the Data

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   (Iterate Steps 2-4 as Necessary)
5. Estimate the treatment effect

Restructuring the Data to Approximate Randomization

Goal: $Z = 0/1$ units that are comparable with respect to $\mathbf{X}$

- Subclassification
  - Stratify units based on $\mathbf{X}$
  - Approximate randomization *within* stratum
- Matching
  - Match each unit to comparison unit(s) based on similarity in $\mathbf{X}$
  - Approximate randomization *within* matches
- Weighting
  - Re-weight each unit based on a function of $\mathbf{X}$
  - Approximate randomization in the *weighted* sample
- Preprocessing/pruning/trimming
  - Identify/exclude units that are *too different* from comparator units in terms of $\mathbf{X}$

Propensity Score Matching: Intuition

- Estimate the propensity score, $\hat{e}(X) \equiv \hat{e}$
- Match each $Z = 1$ unit to 1 (or more) $Z = 0$ units that are "close"
  - I.e., $\hat{e}_i \approx \hat{e}_j$

  $\Rightarrow \mathbf{X}_i \approx \mathbf{X}_j$
- Matched pairs (or groups):
  - Consist of 1 treated and ≥ 1 control unit
  - That are balanced on $\mathbf{X}$

  $\Rightarrow$ Treatment "randomized" within pairs/groups

$\Rightarrow$ Data have been restructured to mimic a (block) randomized study
- If $\mathbf{X}$ has the right confounders
- If the matching has successfully balanced $\mathbf{X}$

Lots of Variations

- 1:1 vs. 1:k vs. ratio matching
  - Match (multiple) treated to (multiple) controls with different ratios
- Calipers
  - Include all matches within a certain distance
  - Exclude matches beyond a certain distance
- Better vs. more matches tradeoff
- Hybrid matching based on $\hat{e}(\mathbf{X}_i)$ and individual components of $\mathbf{X}_i$
- (Stuart 2010) is a key review paper for different matching methods
- Some easy-to-use R packages
  - `library(MatchIt)`
  - `library(optmatch)`
  - `library(Matching)`

5 Steps to Restructure the Data

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5. Estimate the treatment effect

Restructuring $\Leftrightarrow$ "Design"

- Whatever process we follow
  - Matching, subclassification, weighting, etc.
  - Propensity score or otherwise

- We should arrive at an instance of a "design"
  - Where the data have been restructured to (hopefully) approximate the features of a randomized study
  - E.g., balance on confounders

- But need to check whether the design was successful....

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  - E.g., balance on confounders

- But need to check whether the design was successful....

  If not, revisit the restructuring...

Balance and Overlap

- A successful *design* or *restructuring* is one that achieves good balance on $\mathbf{X}$
  - And overlap
- This can (and should) be checked empirically
- Good balance $\Rightarrow$ good approximation of a randomized study
  - (Assuming ignorability)
- Bad balance $\rightarrow$
  - Refine the restructuring
  - Frame expectations for causal inference with appropriate caution

Balance Checking

Compare the distribution of $\mathbf{X}$ across treatment/control groups within (matches, subclasses, weighted samples) that result from the restructuring

- Frequently done with *absolute standardized mean differences* (ASMD)
  - $\frac{|\bar{X_k}^1 - \bar{X_k}^0|}{sd(X_k^1)}$ for $k = 1, ..., p$ covariates
  - ASMD$_k \approx 0 \Rightarrow$ Covariate k is "balaced"
- Common to compare ASMD for every covariate *before* restructuring and *after* restructuring

Iterating Steps 2-4

Achieving adequate balance may require several iterations

- Estimate propensity score $\rightarrow$ matches $\rightarrow$ balance diagnostics
- Repeat until balance is achieved

This is ok **as long as you don't choose your "design" based on which one gives the best treatment effect estimate**

- Don't "peek" at the treatment effect estimate during the design phase
- Don't even look at outcomes at all (cf. (Rubin 2008) comments on "objectivity")
- Evaluate different designs based on information on the covariates, treatments

There may be many different ways to achieve a"good design"

- And they may not all provide equivalent answers for the treatment effect

Chapter 2: Estimating Causal Effects

Chapter 2.2: Analyzing Observational Studies

5 Steps to Restructure the Data

According to (Gelman, Hill, and Vehtari 2021)

1. Estimands and Confounders
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   (Iterate Steps 2-4 as Necessary)
5. **Estimate the treatment effect**

After Satisfactory Restructuring...

Analyze just as you would an experiment!

- Regression
- Comparison of means (across subclasses)
- etc.

One main point of the restructuring/design is that the specifics of the analysis strategy *should not make that big of a difference*

- Just as would be the case in a randomized trial
- Conclusions about causal effects should be *robust* with respect to specific analysis strategy **if the design was successful and assumptions like ignorability are satisfied**

Can My (Spatial) Regression Model Do Causal Inference?

Generic (spatial) regression model

$$Y_i = \beta_Z Z_i + f(\mathbf{X}_i, \beta) + \epsilon_i$$

- $\beta_Z = E[E[Y|Z=1,\mathbf{X}] - E[Y|Z=0,\mathbf{X}]]$
- Ignorability + correct model specification
  - $\beta_Z = \underbrace{E[E[Y|Z=1,\mathbf{X}] - E[Y|Z=0,\mathbf{X}]]}_{\text{Observed Data Comparison}} = \underbrace{E[Y(1)-Y(0)]}_{\text{Causal Estimand}}$
  - Causal estimand *is* a model parameter (coincidence)

But what could go wrong?

Can My (Spatial) Regression Model Do Causal Inference?

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  - Causal estimand *is* a model parameter (coincidence)

But what could go wrong?

- More confounding $\rightarrow$ more covariate imbalance $\rightarrow$ regression has to "work harder" to adjust
  - Amplifies threat of model misspecification

Covariate Balance and Overlap

Generic (spatial) regression model

$$Y_i = \beta_Z Z_i + f(\mathbf{X}_i, \beta) + \epsilon_i$$

- Regression model offloads all confounding adjustment to the parametric specification
- Nothing ensures covariate balance when estimating $E[Y|Z=1,X] - E[Y|Z=0,X]$
- More imbalance $\rightarrow$ more reliance on model specification
- Extreme case: lack of "overlap"
  - When there are not *both* $Z=0$ and $Z=1$ units with $\mathbf{X}=\mathbf{x}$
  - Model doesn't "know this"
  - Regression *extrapolates* expected mean difference in $[Y|Z=0,\mathbf{X}=x]$ and $[Y|Z=1,\mathbf{X}=x]$ *even when there is no such data in the sample*

Causal Inference with (Spatial) Regression Models

- Could still use potential outcomes framework to:
  - Clarify exactly what is meant by "causal effect"
  - Reason about the assignment mechanism $\rightarrow$ identify confounders
- Design/restructuring can mitigate (or at least diagnose) threats to causal validity
  - E.g., diagnose the threat of confounding with balance/overlap checks
- Design as "preprocessing" before regression modeling (Ho et al. 2007)
  1. Construct a (propensity) matched sampled
  2. Fit the regression model

Beyond Matching and Regression

Matching is just one (intuitive) illustration of causal inference methodology!

- Useful for building intuition
- Many other methods
  - Inverse weighting
    - E.g., Marginal structural models (Robins, M. Hernan, and Brumback 2000)
  - Flexible response-surface modeling
    - E.g., Bayesian causal forests (Hahn, Murray, and Carvalho 2020)
  - Double/debiased machine learning
    - E.g., (Kennedy 2024)

Chapter 2: Estimating Causal Effects

Chapter 2.3: Case Study: Effects of Power Plant Emissions Controls

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Adjusting for unmeasured spatial confounding with distance adjusted propensity score matching

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SUMMARY

Propensity score matching is a common tool for adjusting for observed confounding in observational studies, but is known to have limitations in the presence of unmeasured confounding. In many settings, researchers are confronted with spatially-indexed data where the relative locations of the observational units may serve as a useful proxy for unmeasured confounding that varies according to a spatial pattern. We develop a new method, termed distance adjusted propensity score matching (DAPSm) that incorporates information on units' spatial proximity into a propensity score matching procedure. We show that DAPSm can adjust for both observed and some forms of unobserved confounding and evaluate its performance relative to several other reasonable alternatives for incorporating spatial information into propensity score adjustment. The method is motivated by and applied to a comparative effectiveness investigation of power plant emission reduction technologies designed to reduce population exposure to ambient ozone pollution. Ultimately, DAPSm provides a framework for augmenting a "standard" propensity score analysis with information on spatial proximity and provides a transparent and principled way to assess the relative trade-offs of prioritizing observed confounding adjustment versus spatial proximity adjustment.

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Case Study: Power Plant Emissions Controls

Public health burden associated with **exposure to ambient ozone (O_3) pollution**

Emit $NO_x \rightarrow$ interact with weather/atmosphere$\rightarrow$ O_3



Regulations designed to limit population exposure to O_3

- Power plants install a variety of strategies to reduce NO_x
- What is the **causal effect** of SCR/SNCR systems on reducing ambient O_3?

Data for Evaluating SCR/SNCR on Power Plants

- Coal and natural gas power plants during June-August 2004
- 152 facilities "treated" with SCR/SNCR, 321 "controls"
- Outcome: 4th maximum ambient O_3 concentration
- Covariates: Power plant characteristics, demographics, **weather**



Fig. 1. Map of facilities, colored by whether they are treated (yellow) or control (red), and map of ozone concentration surrounding power plants.

(Figure from Papadogeorgou et al. 2019)

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   (Iterate Steps 2–4 as Necessary)

5. Estimate the treatment effect

1. Estimands and Confounders

Let Z_i denote the indicator of whether the ith of n observations is subject to treatment, for example, the indicator of whether a power plant is treated with SCR/SNCR ($Z_i = 1$) or not ($Z_i = 0$). Let Y_i be a continuously-scaled outcome, for example, ambient ozone concentration in the area surrounding power plant i.

Each unit or observation is assumed to have two potential outcomes (Rubin, 1974), one under each value of the treatment. We denote $Y_i(0)$, $Y_i(1)$ as the potential outcome of ambient ozone concentration in the area around unit i under value of $z = 0, 1$ respectively. Assuming that the indexing of the observations is done at random, the index i is suppressed. Interest often lies in the estimation of the average treatment effect in the treated population, defined as $ATT = E[Y(1) - Y(0)|Z = 1]$.

Among the assumptions required to estimate the ATT with observational data is that of "ignorable

1. Estimands and Confounders

We compiled a national data source linking information on power plants, ambient pollution, population demographics, and weather. The resulting data set consists of 473 power generating facilities powered by either coal or natural gas during June, July, and August 2004, which represents the peak ozone season in a year following the institution of important NO_x and ozone regulations. Covariate information ($\mathbf{X}$) on each facility includes power plant operating characteristics such as operating capacity and heat input, as well as area level characteristics such as temperature and population demographics. As a measure of ozone in the area surrounding each power plant (Y), we use the fourth highest daily ozone concentration, averaged across all monitoring locations within a 100 km radius. This measure is chosen to mimic the National Ambient Air Quality Standard for ozone, which is based on the annual fourth-highest daily maximum 8-hour average ozone concentration. Appendix F of the supplementary material available at *Biostatistics* online has a detailed description on the exact construction of the data set used in the final analysis, including references to publicly-available raw data sets and R scripts used for data construction and linkage.

We consider as "treated" the power plants for which at least 50% of the heat input is to facility units with at least one SCR or SNCR technology installed ($Z = 1$, 152 facilities), with the remaining plants regarded as untreated ($Z = 0$, 321 facilities). 67.7% of facilities have either 0% or 100% of their heat input used by units with installed SCR or SNCR control technologies, suggesting robustness to the 50%

$$\text{ATT: } \hat{\tau} = \frac{1}{N_T} \sum_{i \in \mathcal{T}} \left(Y_i - \sum_{j \in \mathcal{C}} W_{ij} Y_j \right), \ \sum_j W_{ij} = 1$$

Data and Code

Harvard Dataverse Link has the data sets used for the analysis

DAPSm Paper Analysis has the code for the analyses reported in the paper

DAPSm Github has the software for implementing the Distance Adjusted Propensity Score Matching procedure we will use later

Useful Literature

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Chapter 3: Spatial Confounding

Chapter 3.1: What is Spatial Confounding?

"Spatial Confounding"

"**Spatial confounding**" used differently across different literatures:

1. In the *spatial statistics* literatures
   - Used to explain why point estimates differ between "spatial" and "non-spatial" regression models (Hodges and Reich 2010; Paciorek 2010; Dupont, Wood, and Augustin 2022)
   - Actually describes several different statistical phenomena
   - Good review: (Khan and Calder 2022)

"Spatial Confounding"

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   - Actually describes several different statistical phenomena
   - Good review: (Khan and Calder 2022)

2. In the *causal inference* literature:
   - Spatial confounding has received comparatively recent consideration
   - Has a more specific meaning

Both literatures are currently active, but we will focus on the latter

Spatial Confounding in **Spatial Statistics**

Existing (regression-based) methods:

- Partially linear models with splines (Dupont, Wood, and Augustin 2022)
- Ordinary least squares
- Gaussian process regression
- Generalized least squares
- Restricted spatial regression (Hodges and Reich 2010)
- Spatially varying coefficient model (SVC) (Gelfand et al. 2003)

Note: These are all specific *modeling* procedures that mix causal assumptions with specifications of the data-generating process

Spatial Confounding in **Spatial Statistics**

Represents several statistical phenomena:

1. Random effect collinearity (Dupont, Wood, and Augustin 2022; Hodges and Reich 2010)
2. Regularization bias (Dupont, Wood, and Augustin 2022)
3. Concurvity (Ramsay, Burnett, and Krewski 2003)
4. Omitted confounder bias

Examples of concepts that mix causal assumptions with features of model specification

Spatial Confounding in **Causal Inference**

- Has explicit definition
- Is not tied to specification of any particular (regression) model
- Expressly related to assumptions about the *assignment mechanism*
  - One formalization of *unobserved confounding*
- Has fairly recently garnered interest
- Could resolve some of the inconsistencies/confusion in the spatial statistics literature

Uses of "Spatial Confounding"

Causal Inference

- Missing spatial information confounds $Z \to Y$
- Identification $>$ modeling
- Solved via designs/analyses that recover/approximate ignorability & positivity

Spatial Statistics

- Concurvity/collinearity between Z and spatial random effects/smoothers
- Regularization shifts signal away from Z
- Modeling bias/variance issue

Goals: A Causal Inference View of Spatial Confounding

- Introduce model-agnostic definition of "spatial confounding" rooted in causal inference
- Pin down what breaks (e.g., ignorability & positivity) in space and what assumptions are needed to "make things work"
- Some practical design/analysis ideas
  - Revisit case study

Notation

- $i = 1, 2, \ldots, n$ observational units
- Exposure, Z_i
- Outcome, Y_i
- Observed covariates: $\mathbf{X}_i$

Notation

- $i = 1, 2, \ldots, n$ observational units
- Exposure, Z_i
- Outcome, Y_i
- Observed covariates: $\mathbf{X}_i$
- Each has location, $s_i \in \mathcal{D} \subset \mathbb{R}^2$
  - E.g., a lat/long
- Unmeasured covariates: U_i

Unconfounded Assignment Mechanisms

An assignment mechanism is *unconfounded* or *ignorable* (conditional on $\mathbf{X}$) if it does not depend on *any* potential outcomes

$$Pr(\mathbf{Z}|\mathbf{X}, \mathbf{Y}(0), \mathbf{Y}(1)) = Pr(\mathbf{Z}|\mathbf{X})$$

$$Z \perp\!\!\!\perp Y(0), Y(1)|X$$

Standard Confounding Assumption

$$Y_i(z), Y_i(z') \perp\!\!\!\perp Z_i | \mathbf{L}_i \tag{1}$$

where $\mathbf{L}_i$ is the set of all confounders required to satisfy (1)

Traditional confounding takes $\mathbf{L}_i = \mathbf{L} = (X_1, X_2, \ldots, X_p)$

- Confounders are the observed $\mathbf{X}$

Standard Confounding Assumption

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where $\mathbf{L}_i$ is the set of all confounders required to satisfy (1)

Traditional confounding takes $\mathbf{L}_i = \mathbf{L} = (X_1, X_2, \ldots, X_p)$

- Confounders are the observed $\mathbf{X}$
- But what about unobserved confounders, U?
- But what about unobserved confounders that we expect to vary spatially?

Unmeasured Confounding

$$Y_i(z), Y_i(z') \perp\!\!\!\perp Z_i | \mathbf{L}_i$$

Unmeasured Confounding

$$Y_i(z), Y_i(z') \perp\!\!\!\perp Z_i | \mathbf{L}_i$$

Traditional spatial confounding takes $\mathbf{L}_i = \mathbf{L} = (\mathbf{X}, U)$

- Confounders are the observed $\mathbf{X}$ and unobserved U

A fundamental fact: in the presence of general unmeasured confounding, the identification and estimation of causal effects (i.e., the complete elimination of omitted variable bias) is not possible

Traditional Spatial Confounding Assumption

$$Y_i(z), Y_i(z') \perp\!\!\!\perp Z_i | \mathbf{L}_i$$

Traditional spatial confounding takes $\mathbf{L}_i = \mathbf{L} = (\mathbf{X}, U)$

- Confounders are the observed $\mathbf{X}$ and unobserved U
- U has a "spatial structure": $U_i = g(s_i)$ with $g(\cdot)$ unknown

Identifiability in the Presence of Spatial Confounding

Key assumptions summarized: had U been observed, causal effects of Z on Y would be identified.

No structure to $U \Rightarrow$ cannot recover causal estimates

Thus, the presence of spatial confounding may be a "blessing" contrasted with unmeasured confounding that has no spatial structure.

$$\Downarrow$$

Key question: how to leverage spatial structure to create a bridge between the observed data and the unmeasured confounder?

Baseline Setting: Identification without Space

Notation

- Y: outcome
- Z: treatment/intervention
- X: measured confounders

Standard ingredients for identifiability:

- SUTVA: no interference or hidden versions of treatment
- Consistency: $Y^{obs} = Y(Z = z)$
- Conditional ignorability/exchangeability: $Y(Z) \perp\!\!\!\perp Z \mid X$
- Positivity: $0 < \Pr(Z = 1 \mid X = x) < 1$

Identifiability in the Presence of Spatial Confounding

Following (Gilbert et al. 2024)

Updated notations:

- Y: outcome
- Z: treatment/intervention
- ~~X: measured confounders~~ for simplicity
- U: (spatially varying) unmeasured confounders

Key assumptions:

- *A1.* Ignorability conditional on unmeasured confounder U:

$$Y(z) \perp\!\!\!\perp Z \mid U$$

- *A2.* Positivity with respect to U:

$$\forall (z,u): \; f_Z(z) > 0 \text{ and } f_U(u) > 0 \implies f_{Z,U}(z,u) > 0.$$

Ignorability in Space

Assumptions and propositions:

- *A3.* All variation in U is captured by S: $U = g(S)$ for a fixed, measurable function g.
- *A4.* Conditioning on S does not induce confounding: $Y(z) \perp\!\!\!\perp Z \mid (S, U)$.
- **Proposition 1**

$$\left.\begin{array}{l} A3.\ U = g(S) \\ A4.\ Y(z) \perp\!\!\!\perp Z \mid (S, U) \end{array}\right\} \implies Y(z) \perp\!\!\!\perp Z \mid S.$$

- *A5. (alt. to A4)* Spatial location only affects the joint distribution of potential outcomes and exposure through the unmeasured confounder: $(Y(z), Z) \perp\!\!\!\perp S \mid U$.

Positivity in Space

Assumptions and propositions:

- *A2.* Positivity with respect to U:

$$\forall(z,u): \;\; f_Z(z) > 0 \text{ and } f_U(u) > 0 \implies f_{Z,U}(z,u) > 0.$$

- *A7.* S is only associated with Z through U: $Z \perp\!\!\!\perp S \mid U$ (can be implied from *A5*).
- **Proposition 2**

$$\left.\begin{array}{l} A2.\text{ Positivity w.r.t. } U \\ A3.\ U = g(S) \\ A7.\ Z \perp\!\!\!\perp S \mid U \end{array}\right\} \implies \text{positivity holds conditional on } S.$$

The Punchline

Adjusting for S can be a "stand in" for adjusting for U if:

- U can be completely explained by S
- S does not have any additional relationship to Z, Y that does not involve U
- Exposure varies for reasons *other* than S
  - Z varies within levels of S

$\Rightarrow$ If U is required to adjust for confounding, can use S to identify causal effects

- No model!
- All identification assumptions
  - Focus on assignment mechanism and the relationship between U/S

Chapter 3: Spatial Confounding

Chapter 3.2: Spatial Confounding Adjustment

Approaches to Estimating Causal Effects

- "Causal inference" methods
  - Inverse weighting by a "spatial propensity score" (Davis et al. 2019)
  - Spatial covariate balancing methods (Woodward, Dominici, and Jose R. Zubizarreta 2025)
  - Double machine learning/doubly robust methods (Gilbert et al. 2024)
  - Spatial matching (Papadogeorgou, Choirat, and Zigler 2019; Keele, Titiunik, and José R. Zubizarreta 2015)

Illustrative Example: "Spatial" Matching

Ignorability:

$$Y_i(z), Y_i(z') \perp\!\!\!\perp Z_i | \mathbf{X}_i, U_i$$

- **Goal of matching**: Match units that are comparable on confounders, $\mathbf{X}, U$
  - E.g., via propensity scores
- **Problem:** U are not observed - how to match on U
- **Solution:** Match units that are spatially close
  - (i, j) spatially close $s_i \approx s_j \leftrightarrow U_i \approx U_j$
  - If $U = g(s_i)$, sufficiently smooth

"Spatial Propensity Scores"

Two perspectives:

1. Estimate a **spatial propensity score**
   - Include some function of latitude/longitude in the propensity score model
   - Include a spatially-varying random effect in the propensity score model (e.g., (Davis et al. 2019))
   - $\hat{e}_i \approx \hat{e}_j$ if (i, j) are spatially close

2. Implement a **spatial matching procedure**
   - A procedure for ensuring that matches are geographically close

"Spatial Propensity Scores"

2. Implement a **spatial matching procedure**
   - A procedure for ensuring that matches are geographically close

"Spatial Propensity Scores"

2. Implement a **spatial matching procedure**
   - A procedure for ensuring that matches are geographically close

   Navigate the tradeoff between:
   - Making matches as similar as possible with respect to X
   - Small distance of matched pairs to capture similarity in U

Distance Adjusted Propensity Score Matching

(Papadogeorgou, Choirat, and Zigler 2019)

For units i, j:

$$DAPS_{ij} = \underbrace{w|\hat{e}_i - \hat{e}_j|}_{\text{Observed Confounding}} + \underbrace{(1-w)Dist_{ij}}_{\text{Geographic Distance}} \quad w \in [0,1]$$

- $\hat{e}_i, \hat{e}_j$ are estimated propensity scores
  - Estimated using only X
- $Dist_{ij}$ is the geographic distance between i, j
- w specifies relative importance of the observed and unobserved confounders
  - High w - most matching weight on observed covariates
  - Low w - most matching weight on spatial proximity
  - w controls the tradeoff between observed confounding adjustment and spatial proximity

Distance Adjusted Propensity Score Matching

(Papadogeorgou, Choirat, and Zigler 2019)

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  - High w - most matching weight on observed covariates
  - Low w - most matching weight on spatial proximity
  - w controls the tradeoff between observed confounding adjustment and spatial proximity

1. Calculate $DAPS_{ij}$ for all (treated, control) pairs
2. Match treated/control units with small $DAPS_{ij}$

Two Decisions

To operationalize DAPSm, decisions must be made about:

1. The distance measure, $Dist_{i,j}$
   - One choice: standardized geodistance
   - $Dist_{ij} = \frac{d_{ij} - \min_{TC} d}{\max_{TC} d - \min_{TC} d}$
   - $d_{ij} \equiv$ geodistance between i, j
   - (other possibilities exist)

2. The value of w
   - w close to 1 $\rightarrow$ matches that are similar on observed X
     - ...but may be dissimilar with regard to unobserved U
   - w close to 0 $\rightarrow$ matches very close spatially, similar on U
     - ... but may induce poor adjustment for X

Choosing w

- Data-driven choice of w
  - Implement matching across a range of values of w
  - For each matched dataset, evaluation **balance** of observed covariates with the absolute standardized difference in means (ASDM)
  - Choose the smallest value of w that maintains balance on observed covariates (e.g., ASDM < 0.15)
  - Get matches as close as possible without sacrificing observed confounding adjustment
- I.e., Restructure $\rightarrow$ diagnose $\rightarrow$ iterate

For the selected w and corresponding matches, estimate:

$$\text{ATT: } \hat{\tau} = \frac{1}{N_T} \sum_{i \in \mathcal{T}} \Big(Y_i - \sum_{j \in \mathcal{C}} W_{ij} Y_j \Big), \; \sum_j W_{ij} = 1$$

Chapter 3: Spatial Confounding

Chapter 3.3: Case Study: Effects of Power Plant Emissions Controls

Case Study: Power Plant Emissions Controls

Public health burden associated with **exposure to ambient ozone (O_3) pollution**

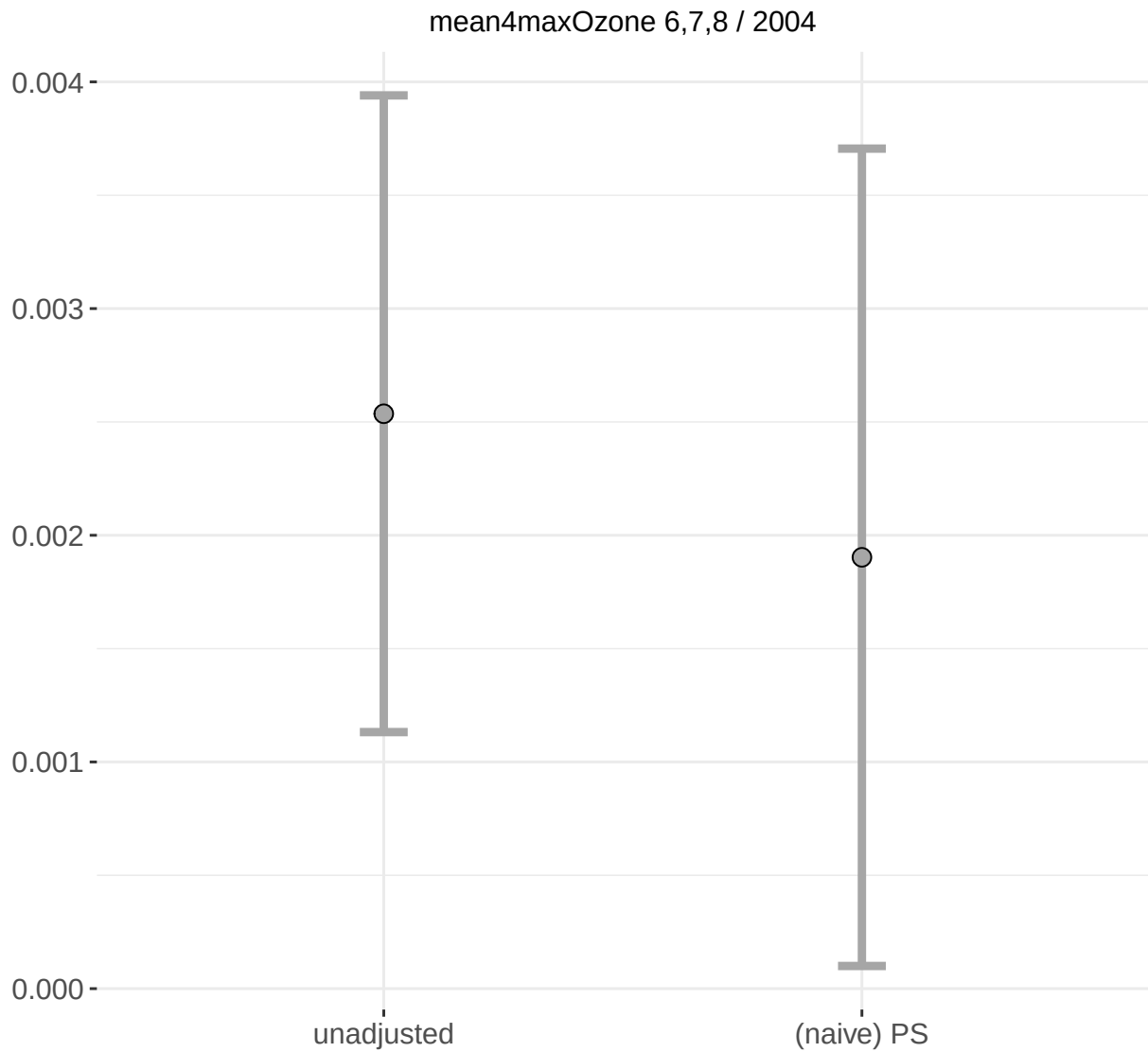
Emit $NO_x \rightarrow$ interact with weather/atmosphere$\rightarrow$ O_3



Regulations designed to limit population exposure to O_3

- Power plants install a variety of strategies to reduce NO_x
- What is the **causal effect** of SCR/SNCR systems on reducing ambient O_3?
  - How do we adjust for the **entire atmospheric (spatial) process of meteorology**?
  - Possibly incompletely measured $\rightarrow U$

Recap Results from Initial Case Study



Recap Results from Initial Case Study

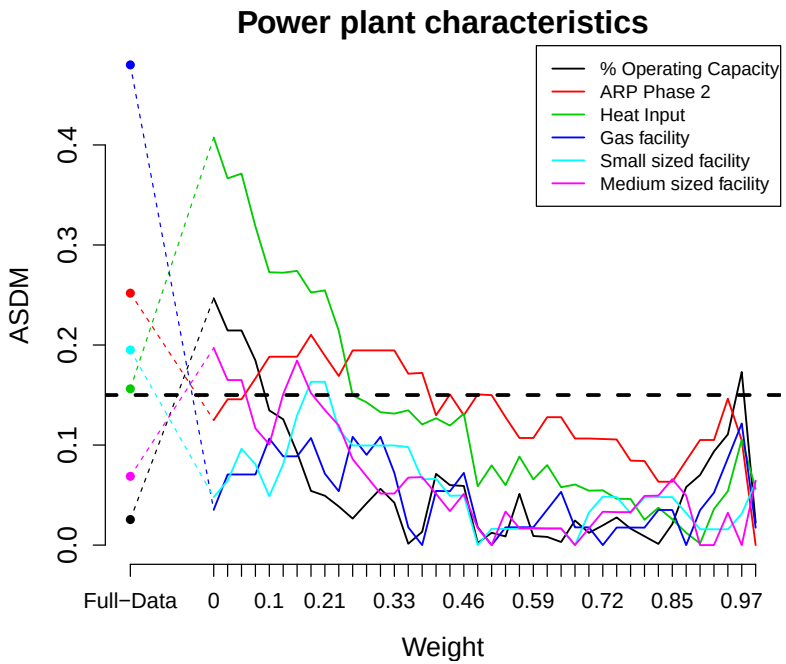


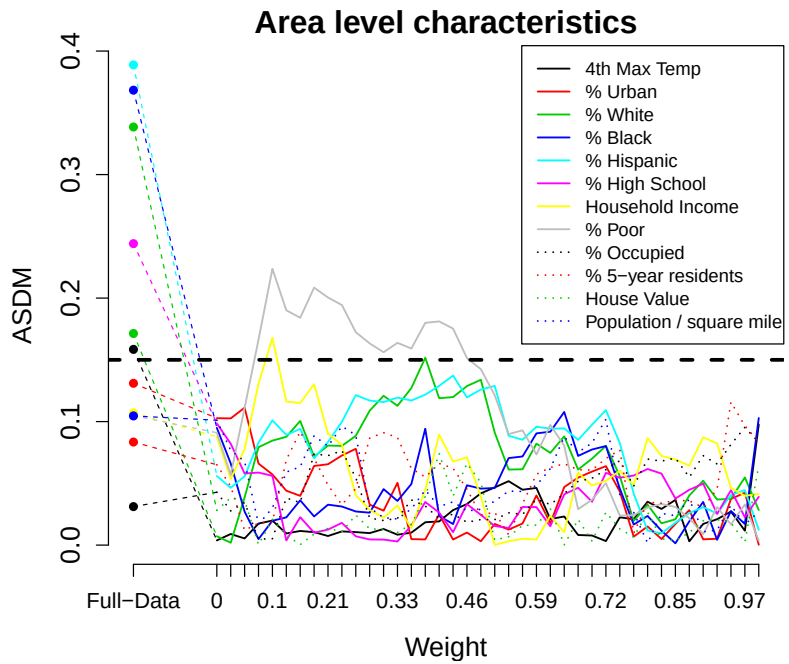
Cut to RStudio to Implement DAPSm

Covariate Balance and Choosing w

$w \to 1 \Rightarrow$ observed covariates

$w \to 0 \Rightarrow$ spatial proximity





w is chosen to be 0.513

Comparing Geographic Matches

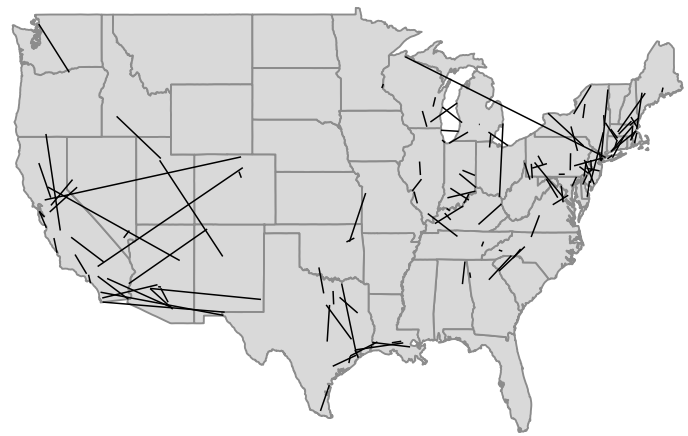
Problem: Complex (unobserved) atmospheric processes confound relationship between SCR/SNCR and O_3 pollution

Solution: Propensity score matching procedure that ensures matches are *spatially close*

Naive pairs

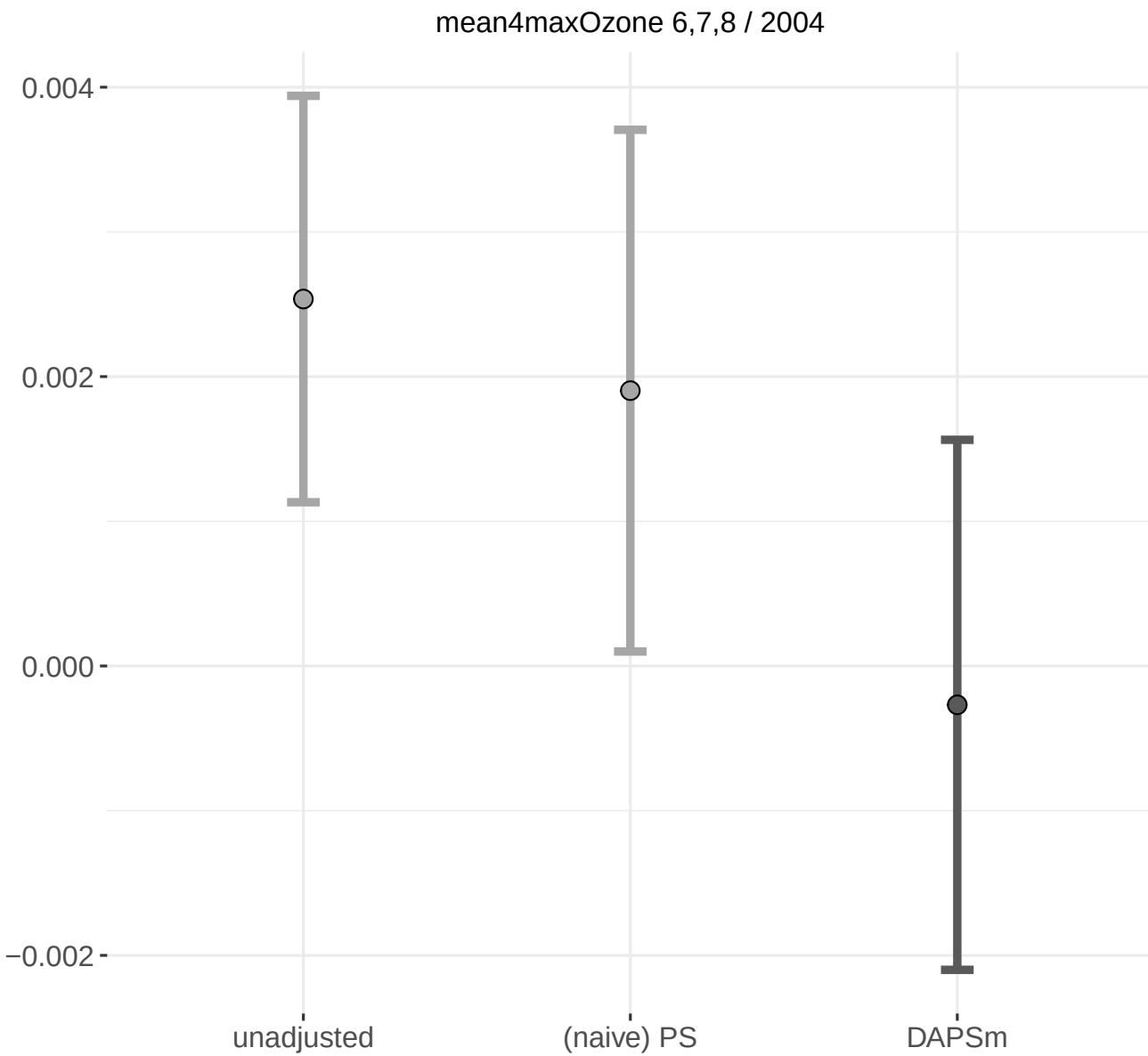


DAPSm pairs



1025mi. vs. 141mi.

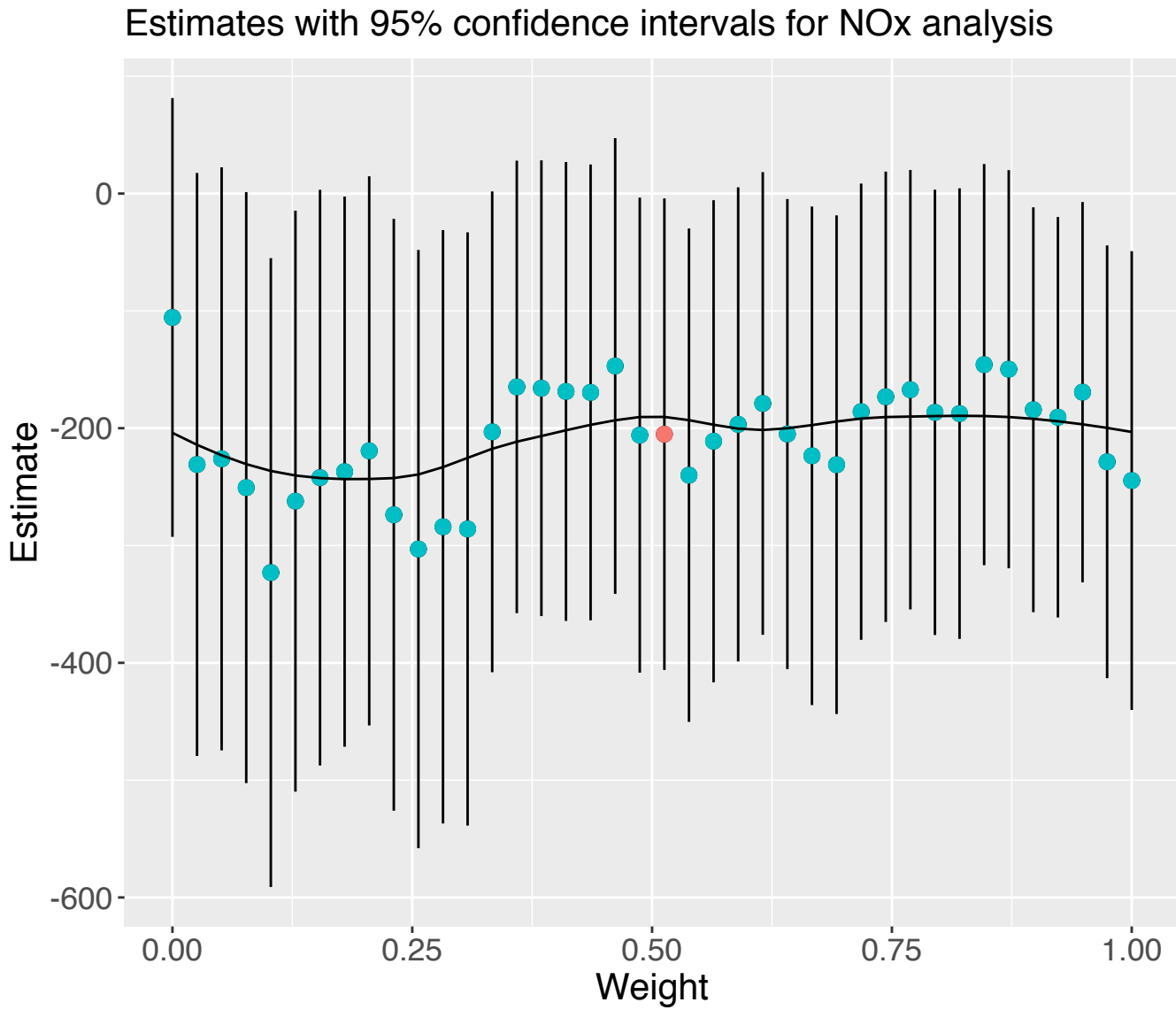
Results for $w = 0.513$



Further evidence of unmeasured spatial confounding

Results across different w

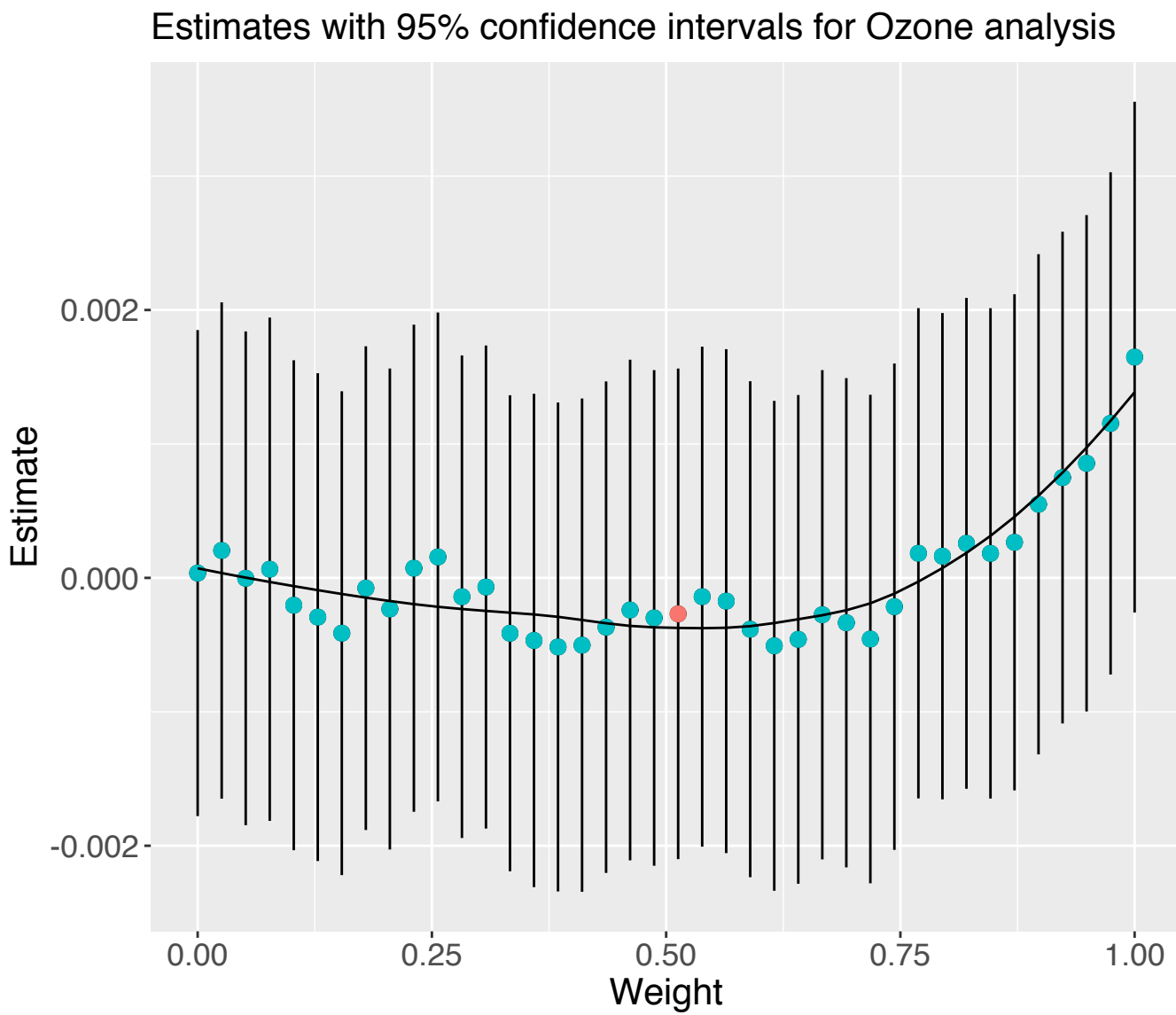
$\mathbf{NO}_x$ Emissions (weak threat of unmeasured spatial confounding)



Further evidence of unmeasured spatial confounding

Results across different w

Ambient O_3 (strong threat of unmeasured spatial confounding)



Useful Literature

Christopher J. Paciorek (Feb. 2010). "The Importance of Scale for Spatial-Confounding Bias and Precision of Spatial Regression Estimators". EN. in: Statistical Science 25.1, pp. 107–125

Brian Gilbert et al. (Oct. 2024). A causal inference framework for spatial confounding. arXiv:2112.14946 [stat.ME]

Emiko Dupont, Simon N. Wood, and Nicole H. Augustin (Dec. 2022). "Spatial+: A Novel Approach to Spatial Confounding". In: Biometrics 78.4, pp. 1279–1290 (+ discussions)

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Gelfand, Alan E et al. (2003). "Spatial Modeling With Spatially Varying Coefficient Processes". In: Journal of the American Statistical Association 98.462, pp. 387–396.

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Hodges, James S. and Brian J. Reich (2010). "Adding Spatially-Correlated Errors Can Mess Up the Fixed Effect You Love". In: The American Statistician 64.4, pp. 325–334.

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Papadogeorgou, Georgia, Christine Choirat, and Corwin M Zigler (2019). "Adjusting for unmeasured spatial confounding with distance adjusted propensity score matching". In: Biostatistics (Oxford, England) 20.2, pp. 256–272.

Ramsay, Timothy, Richard Burnett, and Daniel Krewski (2003). "Exploring bias in a generalized additive model for spatial air pollution data.". In: Environmental Health Perspectives 111.10, pp. 1283–1288.

Woodward, Sophie M., Francesca Dominici, and Jose R. Zubizarreta (2025). Understanding Spatial Regression Models from a Weighting Perspecti

Chapter 4: Frontiers in Spatial Causal Inference

Chapter 4.1: Nonlocal Confounding

General Problem: Adjusting for Weather/Meteorology

Many environmental treatments/exposures are confounded by "weather":

- Routinely associated with treatments/exposures and outcomes
- Complex atmospheric process
- Comprised of many features
- Incompletely observed
- Clear **spatial structure**
- Not characterized by conditions at a single point in space

Confounding process vs. "a confounder"

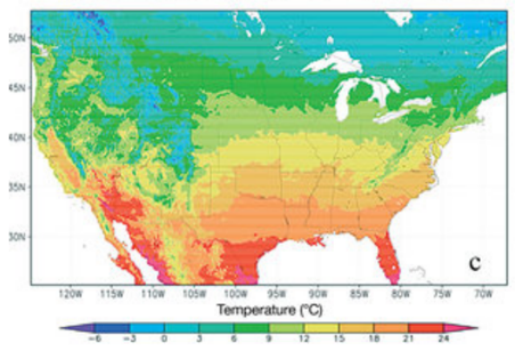
Data Structure: Points + Grids

- $\mathbb{S}$: Set of locations where the outcome and treatment are measured
  - $\mathbb{G} \supset \mathbb{S}$
  - Ex: Point locations of power plants
  - (Binary) treatment: $Z_i \in \{0, 1\}$
  - Outcome: Y_i
  - Note: includes case where $\mathbb{S} \approx \mathbb{G}$



Fig. 1. Map of facilities, colored by whether they are treated (yellow) or control (red), and map of ozone concentration surrounding power plants.
(Figure from Papadogeorgou et al. 2019)

- $\mathbb{G}$: grid over the study region
  - $\boldsymbol{X}_i \in \mathbb{R}^d$: rasterized covariates measured on $\mathbb{G}$
  - Ex: the NARR meteorology data



"Traditional" Spatial Confounding

Standard ignorability assumption:

$$Y_i(z), Y_i(z') \perp\!\!\!\perp Z_i | \mathbf{L}_i$$

Traditional spatial confounding takes $\mathbf{L}_i = (\mathbf{X}_i, U_i)$

- "Standard" unmeasured confounding formulation
- U has a "spatial structure": $U_i = g(s_i)$ with $g(\cdot)$ unknown

Bottom Line: Use some $g(s_i)$ as proxy for U_i

Other Types of "Spatial Confounding"?

Traditional Spatial Confounding:

$$Y_i(z), Y_i(z') \perp\!\!\!\perp Z_i | \mathbf{X}_i, U_i$$

$$\Downarrow$$

$$Y_i(z), Y_i(z') \perp\!\!\!\perp Z_i | \mathbf{X}_i, g(s_i)$$

Other Types of "Spatial Confounding"?

Traditional Spatial Confounding:

$$Y_i(z), Y_i(z') \perp\!\!\!\perp Z_i | \mathbf{X}_i, U_i$$

$$\Downarrow$$

$$Y_i(z), Y_i(z') \perp\!\!\!\perp Z_i | \mathbf{X}_i, g(s_i)$$

Does this capture all types of confounding that arise with spatial data?

Are confounders always features defined on a unit-by-unit basis?

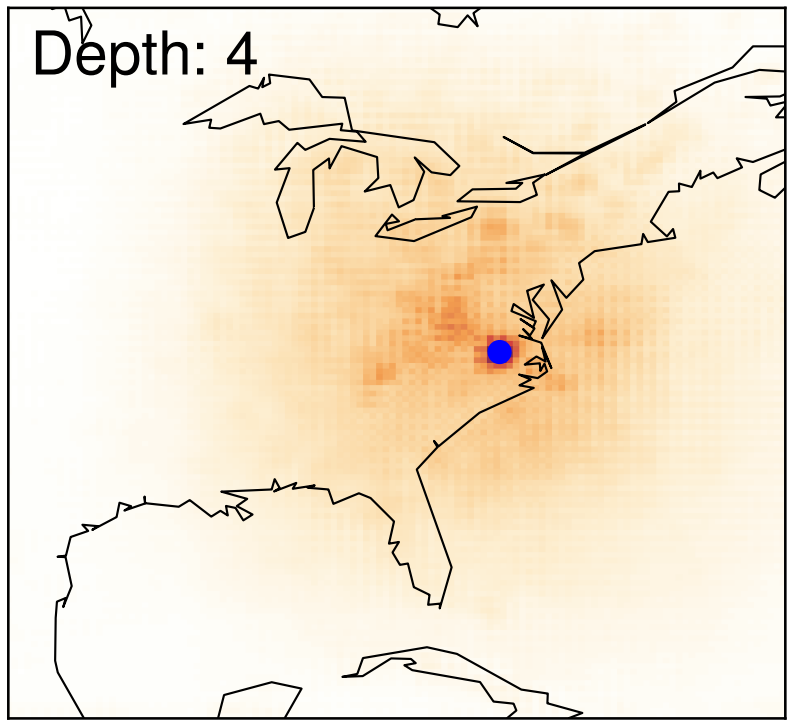
E.g., Can **weather** or other *atmospheric processes* at s_i really be characterized using only features at s_i?

Spatial Nonlocal Confounding

(Tec, Scott, and C. M. Zigler 2023)

- Intervention at a *point* in space (blue dot)
- Gridded atmospheric data as possible *confounders*
  - Meteorology
- Induce confounding both *locally* and by *other* points in space
  - Which points? How strong? (orange grids)

Visualizing structure of **nonlocal confounding**:



(*Receptive field* of a U-net used to learn confounding representation)

Nonlocal Confounding

Definition

$$Y_i(z), Y_i(z') \perp\!\!\!\perp Z_i \mid \boldsymbol{L}_i$$

- Neighborhoods $\{\mathcal{N}_i \subset \mathbb{G} \mid s \in \mathbb{S}\}$ such that $\boldsymbol{L}_i = \mathbf{X}_{\mathcal{N}_i}$
  - $\mathbf{X}_{\mathcal{N}_i} = \{\mathbf{X}_m = (X_{m1}, \ldots, X_{mp}); m \in \mathcal{N}(i)\}$

Nonlocal Confounding

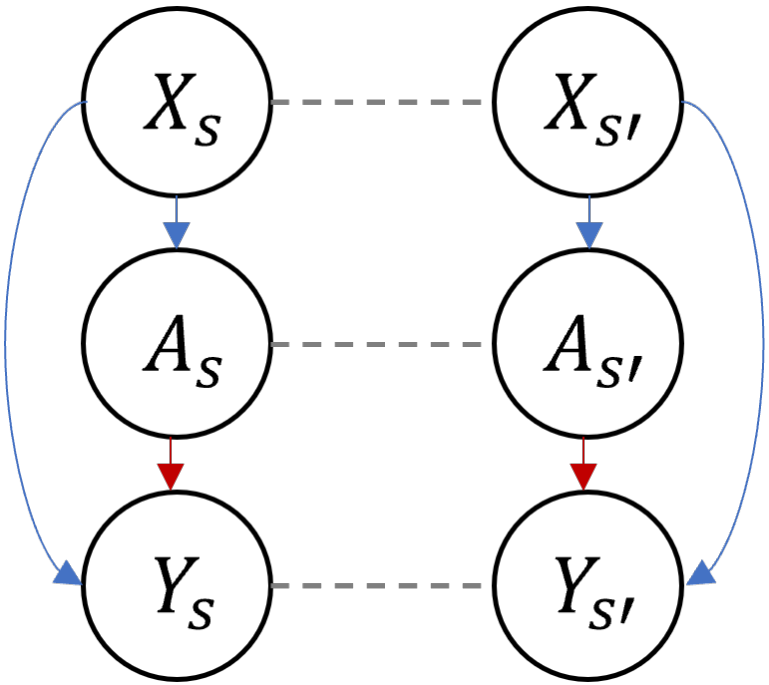
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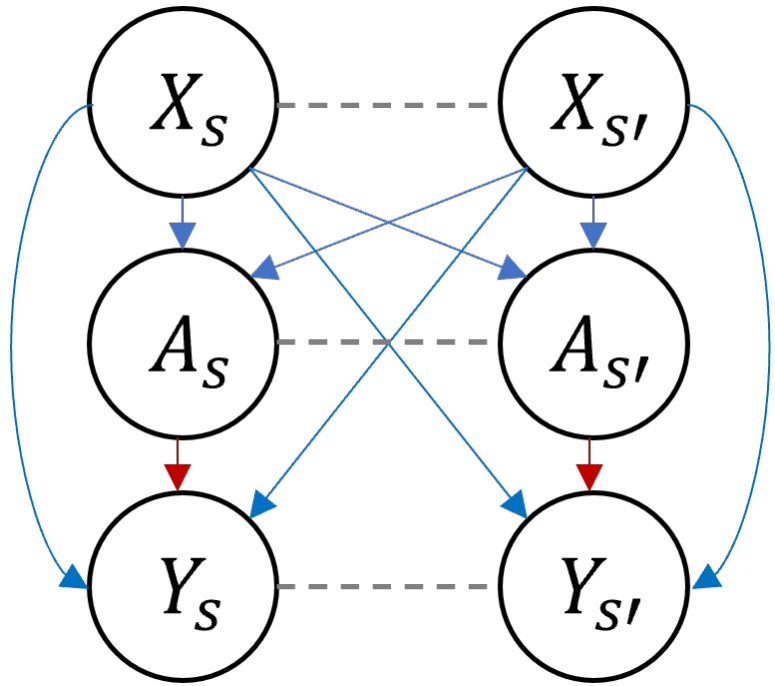
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  - $\mathbf{X}_{\mathcal{N}_i} = \{\mathbf{X}_m = (X_{m1}, \ldots, X_{mp}); m \in \mathcal{N}(i)\}$

- Confounders *for unit* i are:
  1. Observed $\mathbf{X}_i$
  2. Observed $\mathbf{X}$ for *other* units in a "neighborhood" $\mathcal{N}(i)$
- Confounding by *other* units' covariates

Local Confounding



Non-local Confounding (NLC)



Note: For this slide, $A_s \equiv Z_s$ and $s = i$

Note: All assuming *no interference* (SUTVA), but more work on this

How to Adjust for NLC

I.e., How to adjust for *other* units' covariates?

Some (simplistic) options:

How to Adjust for NLC

I.e., How to adjust for *other* units' covariates?

Some (simplistic) options:

- Adjust for covariates of all 1st (or $2^{nd}, 3^{rd}$, ...) degree neighbors of s
  - How many neighbors to include?
  - Could get very high dimensional

How to Adjust for NLC

I.e., How to adjust for *other* units' covariates?

Some (simplistic) options:

- Adjust for covariates of all 1st (or $2^{nd}, 3^{rd}$, ...) degree neighbors of s
  - How many neighbors to include?
  - Could get very high dimensional
- Create some simple summary quantity of the neighborhood features
  - E.g., average temperature of all locations within 300km
  - Hard to know what the summary should be

weather2vec (Tec, Scott, and C. M. Zigler 2023)

Nonlocal confounding adjustment with *representation learning* (U-nets)

Convolutional NN architecture that:

- Localizes information from large neighborhoods
- Extracts high-level spatial features using nested convolutions
- Is spatially stationary (i.e., output depends only on covariates used as inputs)
- Transforms grids $\rightarrow$ grids
  - $\mathbf{X}_{\mathbb{G}} \rightarrow \mathbf{R}_{\theta,\mathbb{G}} = f_\theta(\mathbf{X}_{\mathbb{G}})$ of same spatial dimension
  - Unknown θ dictate what non-local information is summarized by $\mathbf{R}_{\theta,\mathbb{G}}$
  - *Supervised* and *semi-supervised* versions

  $\rightarrow$ Use $\mathbf{R}_{\theta,\mathbb{G}}$ for "design"
    - E.g., for inverse propensity weighting

Chapter 4: Frontiers in Spatial Causal Inference

Chapter 4.2: Spatial Interference and Spillover

Treatments/Exposures are Physical Processes

E.g., transport of pollution from sources $\rightarrow$ populations

- Emissions originating at a power plant (**red dot**) *move*
  - Long distances towards secondary formation of ambient $PM_{2.5}$
- Intervening at a power plant impacts health over a *large spatial area*
- $\Rightarrow$ inference on a "network" of interconnected power plants/population locations
  - $\rightarrow$ Causal inference with **interference**

"Standard" Formulation of Potential Outcomes

(without interference)

For each $i = 1, 2, \ldots, N$ define the *potential outcome* that would occur under each possible treatment:

$$Y_i(Z_i = 0), Y_i(Z_i = 1)$$

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Implicitly relies on the Stable Unit Treatment Value Assumption (SUTVA)

- Outcome for j only depends on treatment for j
- Exactly 2 potential outcomes for each j

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- Exactly 2 potential outcomes for each j

Definition of causal effects:

$$Y_i(Z_i = 1) - Y_i(Z_i = 0) \text{ (individual causal effect)}$$

$$E[Y_i(Z_i = 1) - Y_i(Z_i = 0)] \text{ (average causal effects)}$$

Stable Unit Treatment Value Assumption (SUTVA)

SUTVA Part I: No "Interference"

The potential outcome for the ith unit does not depend on the treatment assigned to *other* units. Let $\mathbf{Z} = (Z_1, Z_2, \ldots, Z_N)$.

$$Y_i(\mathbf{Z}) = Y_i(Z_i)$$

This is precisely the assumption that gives us (with a binary treatment) only 2 potential outcomes for each unit

No STUVA $\rightarrow>>$ 2 potential outcomes for each i

Potential Outcomes with Interference

"Standard" Setting: No Interference

Stable Unit Treatment Value Assumption (SUTVA):

$$Y_i(\mathbf{Z}) = Y_i(Z_i)$$

$$E[Y_i(Z_i = 1) - Y_i(Z_i = 0)]$$

Potential Outcomes with Interference

"Standard" Setting: No Interference

Stable Unit Treatment Value Assumption (SUTVA):

$$Y_i(\mathbf{Z}) = Y_i(Z_i)$$

$$E[Y_i(Z_i = 1) - Y_i(Z_i = 0)]$$

Interference $\Rightarrow$ ~~SUTVA~~

- Potential outcome for j depends on $\mathbf{A}$ at *all other* units
  - $Y_i(\mathbf{Z}) = Y_i(Z_1 = Z_1, Z_2 = Z_2, \ldots, Z_N = Z_N)$
  - $\mathcal{Z}(N) = \{0, 1\}^N$: space of possible allocations
  - 2^N unique potential outcomes for each i
  - Vastly complicates causal inference

Example: Two Units, No SUTVA

$Z = (0, 1)$ for receiving (placebo, vaccine). For $N = 2$ units, $\mathbf{Z} = (Z_1, Z_2)$

$\mathbf{Z}$ takes 2^N values: $(0,0), (1,0), (0,1), (1,1)$

Unit	Potential Outcomes			
1	$Y_1(0,0)$	$Y_1(1,0)$	$Y_1(0,1)$	$Y_1(1,1)$
2	$Y_2(0,0)$	$Y_2(1,0)$	$Y_2(0,1)$	$Y_2(1,1)$

Each unit's potential outcome is indexed by *both* units' treatments

Interference $\rightarrow$ New Causal Quantities

Typically, we ask questions such as: "What is the (average) effect of a treatment on a unit's outcome?"

Interference $\rightarrow$ New Causal Quantities

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With **interference**,

(a) impossible to answer such questions without knowing what treatments are applied to *others* that are connected in some way

(b) new causal quantities may be of scientific interest

Interference $\rightarrow$ New Causal Quantities

Typically, we ask questions such as: "What is the (average) effect of a treatment on a unit's outcome?"

With **interference**,

(a) impossible to answer such questions without knowing what treatments are applied to *others* that are connected in some way

(b) new causal quantities may be of scientific interest

- More than just accounting for or addressing correlation among units
- Two different possible goals:
  - "Adjusting" or "accounting" for the interference to get accurate estimates of familiar quantities
  - Estimating new quantities that only make sense in the presence of interference

Ingredients to Causal Inference with Interference

1. Specify something about the structure of interactions among units
   - Which units interfere with which other units?
   - Frequently described as which units are "neighboring" in a network structure
2. Specify something about how the interference "flows" across units
   - Exposure mapping
3. Define new causal estimands
   - No single "average causal effect" of interest
   - Direct, indirect, total, overall effects
4. Estimate causal effects

Key Quantity: Adjacency Matrix

AKA, *interference mapping*

Network interference structure characterized by a matrix $T =$

$$
\begin{array}{c}
 \\
unit_1 \\
\vdots \\
unit_i \\
\vdots \\
unit_N
\end{array}
\begin{array}{c}
\begin{array}{cccccc}
unit_1 & \dots & unit_j & \dots & unit_n
\end{array} \\
\left(\begin{array}{cccccc}
t_{11} & \dots & t_{12} & \dots & t_{1N} \\
\vdots & \dots & \vdots & \dots & \vdots \\
t_{i1} & \dots & t_{ij} = (\text{influence of } unit_i \rightarrow unit_j) & \dots & t_{in} \\
\vdots & \dots & \vdots & \dots & \vdots \\
t_{N1} & \dots & t_{Nj} & \dots & t_{NN}
\end{array}\right)
\end{array}
$$

- $t_{ij} = 1 \Rightarrow$ unit i's outcome depends on unit j's treatment
  - Units (i, j) *interfere*

Interference in Environmental Science

A different type of interference....

- Most common settings: Social networks
  - Unit-to-unit outcome dependencies
  - Infectious diseases, social networks
  - Structure is treated as given, fixed
    - E.g., collect data on connections between units

- Spatial environmental studies
  - Complex exposure dependencies
  - Treatment diffusion
  - Spatial proximity
  - Physical processes
  - Structure is unknown, uncertain, stochastic
    - Must be modeled/estimated

Network Formulation

E.g., (Forastiere, Airoldi, and Mealli 2021)

Formulate observational units as nodes in a network:

- Observational units are *nodes/vertices* in the network/graph ($\mathcal{V}$)
- *Edges* in the network describe which units interfere (T)

$\rightarrow$ Network/Graph: $(\mathcal{V},]T)$

- Examples of an edge between nodes
  - Two people are friends
  - Two locations are geographically close
  - One location is downstream from another
  - ...

Individual and Neighborhood Treatments

In full generality, every unit interference with every other unit:

$\Rightarrow$ $Y_i(\mathbf{Z}) \neq Y_i(\mathbf{Z}')$ for any $\mathbf{Z} \neq \mathbf{Z}'$

Typically, more structure is useful:

- Call $\mathcal{N}_i = \{j : Z_{ij} \neq 0\}$ the *neighborhood* of unit i
  - All of the units that interfere with unit i
- $\mathbf{Z}_{\mathcal{N}_i}$ is the *neighborhood treatment vector* for i
- $Y_i(\mathbf{Z}) = Y_i(Z_i, \mathbf{Z}_{\mathcal{N}_i})$
  - Potential outcomes depend only a unit's *individual* treatment and *neighborhood* treatment
  - Adaptation of SUTVA

Exposure Mappings

Neighborhood structure offers some simplification:

$\Rightarrow$ ~~$Y_i(\mathbf{Z}) \neq Y_i(\mathbf{Z}')$ for any i~~

$\Rightarrow$ $Y_i(\mathbf{Z}) = Y_i(\mathbf{Z}')$ whenever $Z_i = Z_i'$ and $\mathbf{Z}_{\mathcal{N}_i} = \mathbf{Z}'_{\mathcal{N}_i}$

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Even more structure via an *exposure mapping:*

- $\mathbf{G} = (G_1, \ldots, G_N) \equiv g : \mathcal{Z} \times T \to \mathcal{G}$,
  - $\mathcal{Z}$: space of possible $\mathbf{Z}$
  - T is the network adjacency matrix,
  - $\mathcal{G}$ are the possible unit-level exposures
- G_i is a summary of neighbors' treatments
  - E.g., the proportion of treated neighbors
- (Z_i, G_i) can be viewed as a binary treatment with an *individual* component and a *neighborhood* component

Exposure Mappings

Neighborhood structure offers some simplification:

$\Rightarrow$ ~~$Y_i(\mathbf{Z}) \neq Y_i(\mathbf{Z}')$ for any i~~

$\Rightarrow$ ~~$Y_i(\mathbf{Z}) = Y_i(\mathbf{Z}')$ whenever $Z_i = Z_i'$ and $\mathbf{Z}_{\mathcal{N}_i} = \mathbf{Z}'_{\mathcal{N}_i}$~~

$\Rightarrow$ $Y_i(\mathbf{Z}) = Y_i(\mathbf{Z}')$ whenever $Z_i = Z_i'$ and $G_i = G_i'$

Further adaptation of SUTVA

- Treatment/Interference "flows through" the exposure mapping, G
- Following (Forastiere, Airoldi, and Mealli 2021; C. Zigler et al. 2025; Aronow and Samii 2017) and others.

Causal Inference with Interference

Z_i: Treatment for i^{th} unit
Y_i: Outcome for i^{th} unit

Standard Formulation

Stable Unit Treatment Value Assumption
(no interference)

$$\underbrace{Y_i(Z_i = 1)}_{\text{Potential outcome if } Z_i = 1} \quad \text{vs.} \quad \underbrace{Y_i(Z_i = 0)}_{\text{Potential outcome if } Z_i = 0}$$

Exactly 2 potential outcomes per unit

Causal Inference with Interference

Z_i: Treatment for i^{th} unit
Y_i: Outcome for i^{th} unit

Interference Formulation

~~Stable Unit Treatment Value Assumption~~
(general interference)

$$Y_i(\underbrace{Z_1, Z_2, \ldots, Z_N}_{\text{All } N \text{ units}}) \quad \text{vs.} \quad Y_i(\underbrace{Z_1, Z_2, \ldots, Z_N}_{\text{All } N \text{ units}})$$

2^N potential outcomes for each unit!

Bipartite Causal Inference with Interference

Z_i: Treatment for i^{th} unit
Y_i: Outcome for i^{th} unit

Interference Formulation

~~Stable Unit Treatment Value Assumption~~
(interference via *exposure mapping*)

$$Y_i(\quad \underbrace{g(Z_1, Z_2, \ldots, Z_N; T)}_{\text{Function of interference structure}} \quad) \quad \text{vs.} \quad Y_i(\quad \underbrace{g(Z_1, Z_2, \ldots, Z_N; T)}_{\text{Function of interference structure}} \quad)$$

Potential outcomes as function of specified interference structure, encoded with function $g(\cdot)$ and matrix T

Causal Estimands under Interference

... not simply a problem that needs to be addressed...gives rise to new causal quantities of interest' - (Tchetgen Tchetgen and VanderWeele 2012)

- Direct Effects
- Indirect Effects
- Overall Effects
- Total Effects

Direct Effects

Z_i: Treatment assigned to unit i
$g(\mathbf{Z}_{\mathcal{N}_i})$ (function of) treatments assigned i's neighbors

Direct Effects

Z_i: Treatment assigned to unit i
$g(\mathbf{Z}_{\mathcal{N}_i})$ (function of) treatments assigned i's neighbors

$$Y_i(Z_i = 1, g(\mathbf{Z}_{\mathcal{N}_i}) = g) - Y_i(Z_i = 0, g(\mathbf{Z}_{\mathcal{N}_i}) = g)$$

- Effect of treatment (vs. control) on unit i, under a given treatment vector to the other units ($\mathbf{Z}$)
- Could be different for all possible values of $g(\mathbf{Z}_{\mathcal{N}_i}) = g$

Indirect Effects

Z_i: Treatment assigned to unit i
$g(\mathbf{Z}_{\mathcal{N}_i})$ (function of) treatments assigned i's neighbors

$$Y_i(Z_i = 0, g(\mathbf{Z}_{\mathcal{N}_i}) = g) - Y_i(Z_i = 0, g(\mathbf{Z}_{\mathcal{N}_i}) = g')$$

- Could be defined analogously for $Z_i = 1$
- Effect of changing the treatments of other units, holding fixed unit i's treatment

Reducing the Number of Potential Outcomes

Fully general: $Y_i(\mathbf{Z} = (Z_1, Z_2, \ldots Z_N)$

- 2^N defined for i

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$$\downarrow$$

Individual/Neighbors $Y_i(Z_i = z, \mathbf{Z}_{\mathcal{N}_i} = \mathbf{Z})$

- $2^{|\mathcal{N}|}$ defined for every i

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- $2^{|\mathcal{N}|}$ defined for every i

$\downarrow$

Exposure Mapping: $Y_i(Z_i, g(\mathbf{Z}_{\mathcal{N}_i}) = G_i)$

- $2 \times$ [levels of G] defined for every i

Reducing the Number of Potential Outcomes

Fully general: $Y_i(\mathbf{Z} = (Z_1, Z_2, \ldots Z_N)$

- 2^N defined for i

$\downarrow$

Individual/Neighbors $Y_i(Z_i = z, \mathbf{Z}_{\mathcal{N}_i} = \mathbf{Z})$

- $2^{|\mathcal{N}|}$ defined for every i

$\downarrow$

Exposure Mapping: $Y_i(Z_i, g(\mathbf{Z}_{\mathcal{N}_i}) = G_i)$

- $2 \times$ [levels of G] defined for every i

$\downarrow$

Causal estimands based on (Z_i, G_i)

Example: Interventions on Coal Power Plants
(Wikle and C. M. Zigler 2024)



- Some install "scrubbers" in response to regulations

$\rightarrow$ Reduce SO_2 emissions

$\rightarrow$ Reduce ambient pollution
  - Reducing secondary formation of $PM_{2.5}$ (SO_4^{2-})

$\rightarrow$ Improve health

Question: How do scrubbers on coal-fired power plants causally affect pediatric asthma emergency department (ED) visits in Texas?

Central Challenge: Pollution Transport

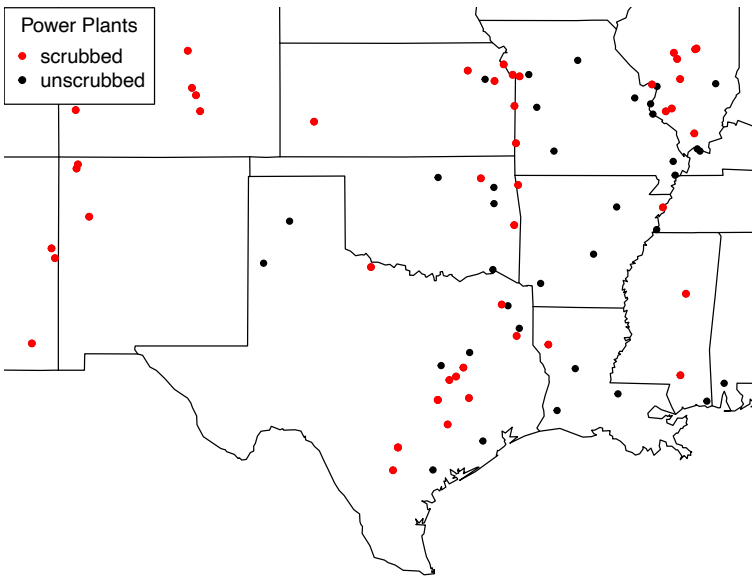
Movement of pollution from sources $\rightarrow$ populations

- Emissions originating at a power plant (**red dot**) *move*
  - Long distances towards secondary formation of ambient $PM_{2.5}$
- Intervening at a power plant impacts health over a *large spatial area*
- $\Rightarrow$ inference on a "network" of interconnected power plants/population locations
  - $\rightarrow$ Causal inference with **interference**

Intervention Effects Propagate Across a Network

Network of $\sim$100 power plants and $\sim$ 2000 population locations

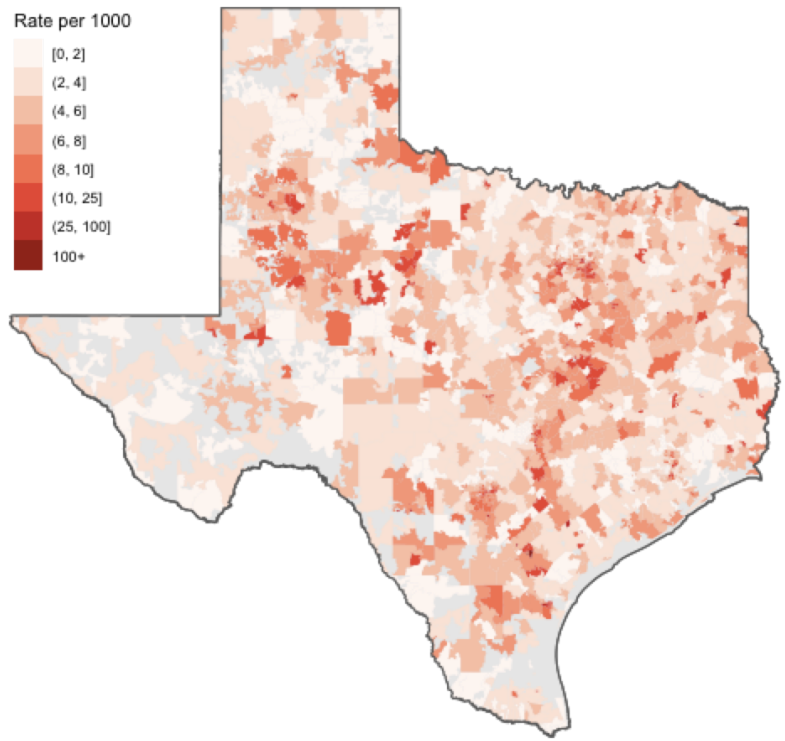
Interventions at Power Plants:
"Scrubber" Installation



Affect

$\longrightarrow$

Health at Population Locations
(ZIP codes)



Outline of Methods Development

1. Formalize potential outcomes, causal estimands
   - Potential outcomes on bipartite graphs
   - "Upwind" and "local" causal effects

2. Characterize the structure of interference
   - Physical/chemical process of pollution transport
   - Statistical (vs. engineering approach)

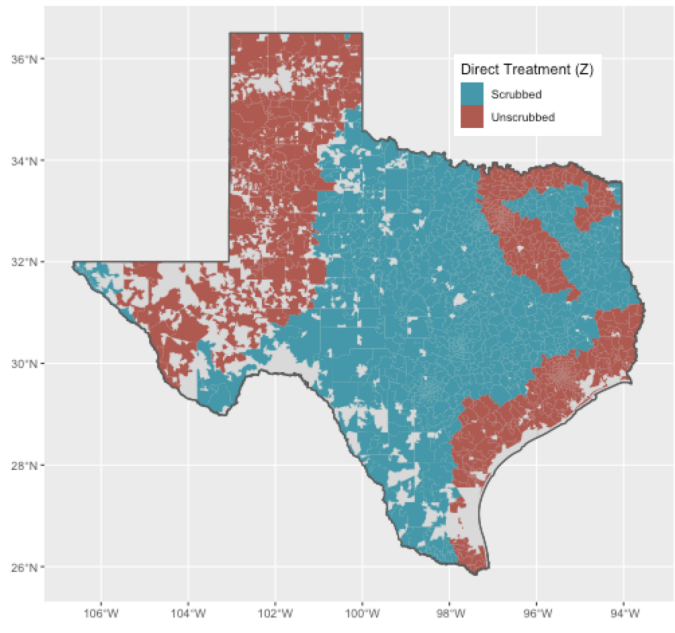
3. Estimate causal effects
   - Bayesian nonparametric regression trees
   - Propagate interference uncertainty
   - Administrative health data: Medicare, Texas Health Care Information Collection
     - All-cause mortality, asthma ED visits

Formulate a "bivariate treatment":

Recall interference map: $T_\theta = \{t_1, t_2, \ldots, t_n\}$

(1) "Individual"-level treatment of the *key associated* power plant, Z_i

- Scrubber status of the plant closest to location i

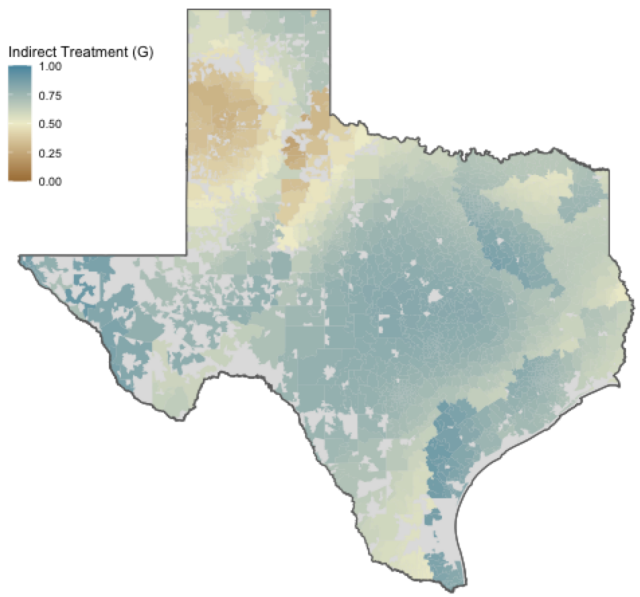


Formulate a "bivariate treatment":

Recall interference map: $T_\theta = \{t_1, t_2, \ldots, t_n\}$

(2) "Neighborhood"-level treatment of the "upwind" power plants

- "Exposure Mapping:" $g_i(\cdot; T_\theta) : \{0,1\}^{J-1} \to \mathcal{G}_i$
- Weighted proportion of upwind facilities that have scrubbers

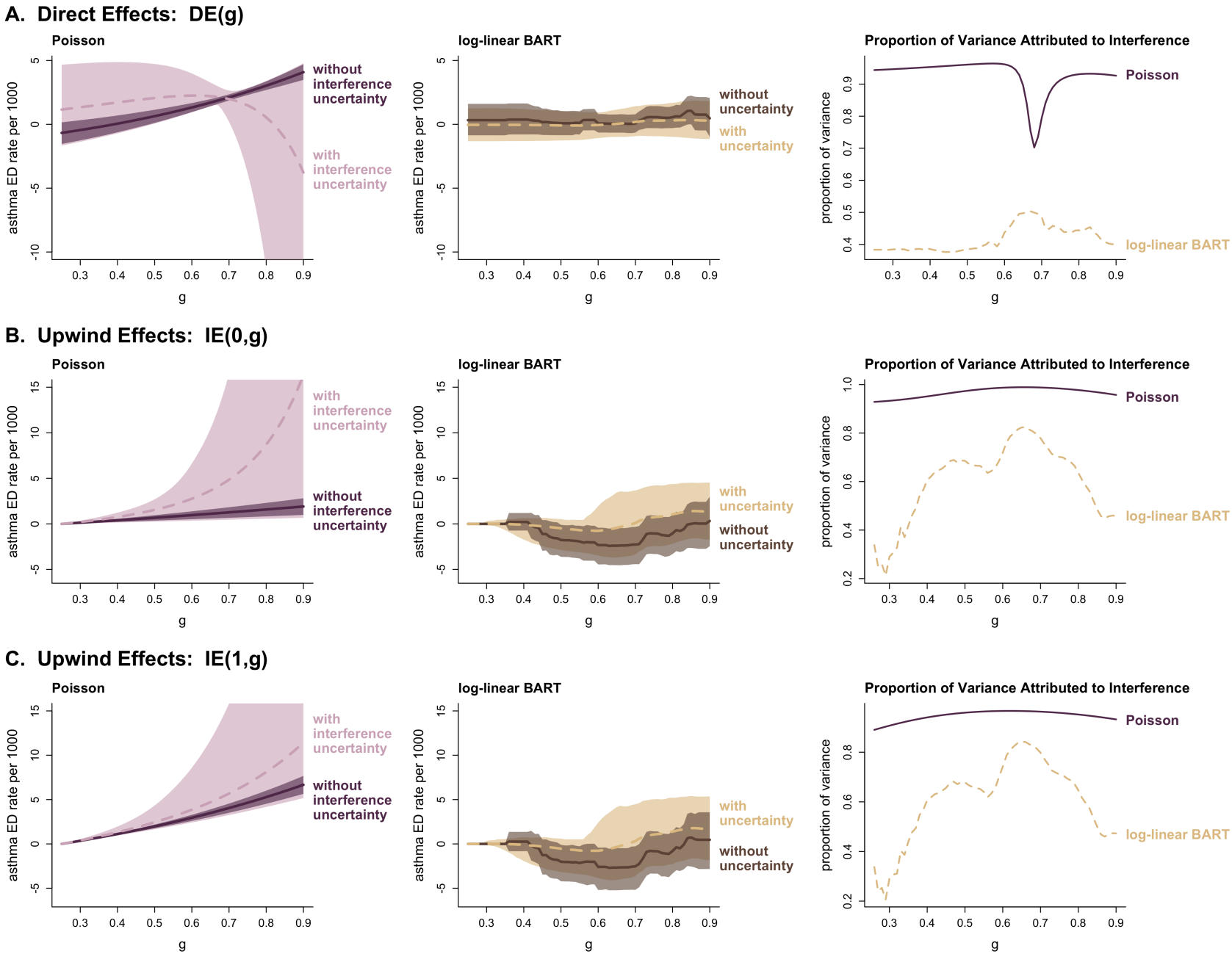


"Upwind Interference" Assumption
(Adaptation of SUTVA)

$$Y_i(\mathbf{Z}) = Y_i(Z_i, G_i) = Y_i(Z_i', G_i') = Y_i(\mathbf{Z}')$$

- All of the interference "flows through" the individual level treatment, Z_i, and the upwind summary quantity, G_i
- **Key interference assumption**

Pediatric Asthma ED



Frontiers in Spatial Causal Inference

- Nonlocal confounding
- Interference/spillover
- Spatial experiments
  - (Wang et al. 2025)
- Spatiotemporal process treatments
  - (Papadogeorgou et al. 2022)

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Chapter 1: Introduction to Causal Inference
- Chapter 1.1: What is Causal Inference?
- Chapter 1.2: The Rubin Causal Model

Chapter 2: Estimating Causal Effects
- Chapter 2.1: "Designing" Observational Studies
- Chapter 2.2: Analyzing Observational Studies
- Chapter 2.3: Case Study: Effects of Power Plant Emissions Controls

Chapter 3: Spatial Confounding
- Chapter 3.1: What is Spatial Confounding?
- Chapter 3.2: Spatial Confounding Adjustment
- Chapter 3.3: Case Study: Effects of Power Plant Emissions Controls

Chapter 4: Frontiers in Spatial Causal Inference
- Chapter 4.1: Nonlocal Confounding
- Chapter 4.2: Spatial Interference and Spillover

Different Emphasis of Spatial Statistics and Causal Inference

Broad Generalization

Spatial Statistics

- Complex regression models
- Encode spatial process or correlation
- Models represent mechanistic understanding of data generating process
- Model parameters $\Leftrightarrow$ "effects" of certain features
- Consider how modeling choices impact parameter estimates

Causal Inference

- Formalize potential outcomes, causal effects
- Explicate assumptions needed for causal interpretation
- Identifying assumptions for causal inference
- Statistical model not required